

COMP5152

Advanced Data Analytics

Linear Regression and Time Series Forecasting

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Correlation and Linear Regression

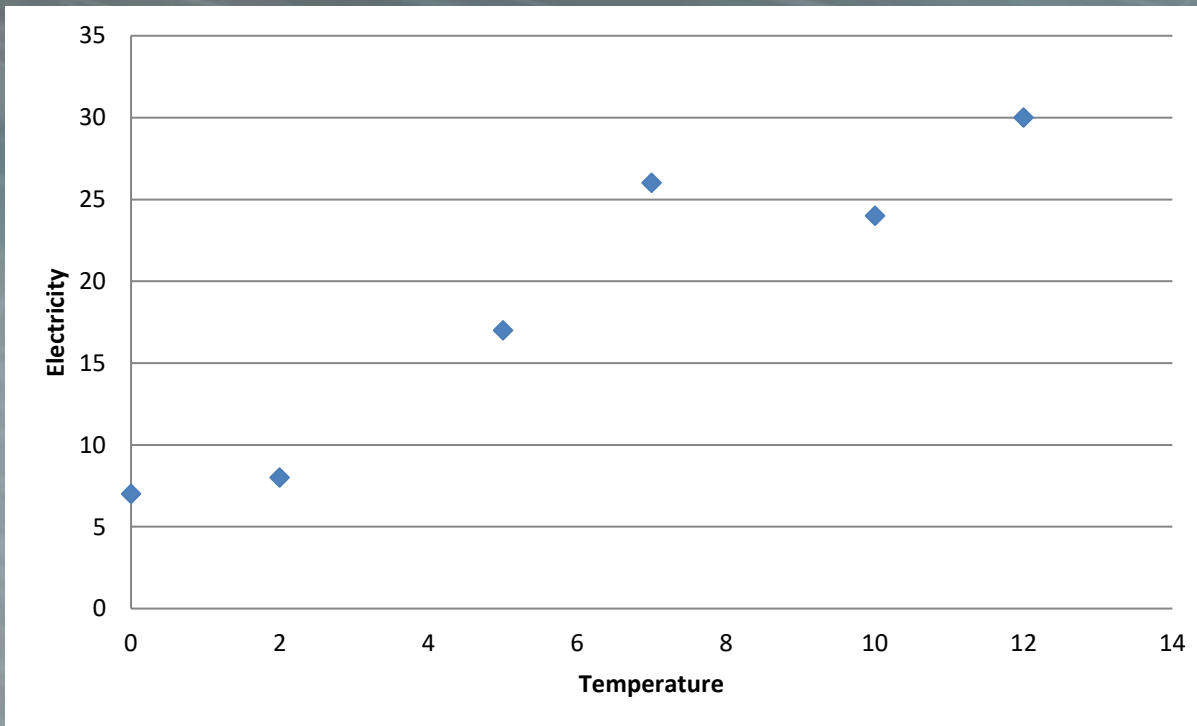
Relationship between Data

- ▶ In many cases, after we have collected large number of data, we are interested to see the relationship between the variables
- ▶ For example, if we collected the consumption of electricity in a region, together with the temperature at the time

Temperature	0	2	5	7	10	12
Electricity	7	8	17	26	24	30

Relationship between Data

- Usually, we plot the data in a scatterplot



- We can see that when **temperature increases**, the **electricity consumption increases too!!**

Temperature	0	2	5	7	10	12
Electricity	7	8	17	26	24	30

Correlation

- ▶ Correlation is help us to check if there is relationships between two or more of our variables.
 - ▶ also the strength of the relationship.

DEFINITION OF CORRELATION

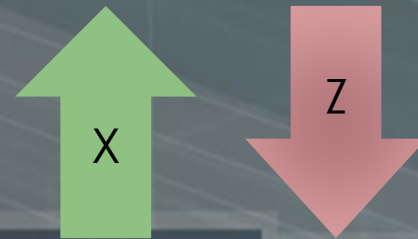
- ▶ “If two or more quantities vary in sympathy so that movements in one tend to be accompanied by corresponding movements in other(s) then they are said to be correlated.”

Types of Correlation

- ▶ Positive and Negative Correlation
- ▶ Simple, Partial and Multiple Correlation
- ▶ Linear and Non-linear Correlation

Positive and Negative Correlation

- ▶ The correlation between two variables is said to be positive or direct if an increase (or a decrease) in one variable corresponds to an increase (or a decrease) in the other.
- ▶ The correlation between two variables is said to be negative or inverse if an increase (or a decrease) corresponds to a decrease (or an increase) in the other.



Simple, Partial and Multiple Correlation

- ▶ **Simple Correlation:** It involves the study of only two variables. For example, when we study the correlation between the price and demand of a product, it is a problem of simple correlation.
- ▶ **Partial Correlation:** It involves the study of three or more variables, but considers only two variables to be influencing each other. For example, if we consider three variables, namely yield of wheat, amount of rainfall and amount of fertilizers and limit our correlation analysis to yield and rainfall, with the effect of fertilizers removed, it becomes a problem relating to partial correlation only.

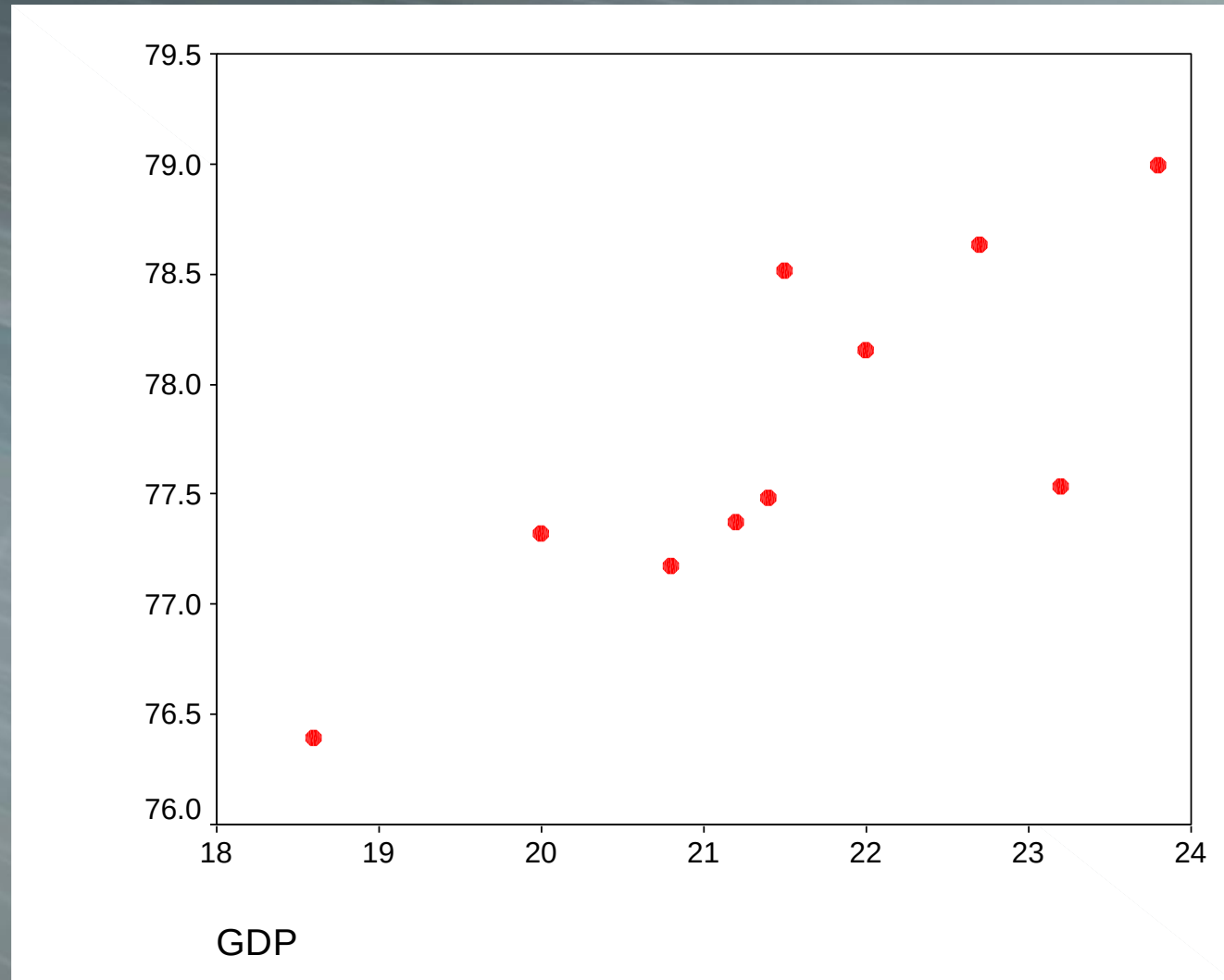
Simple, Partial and Multiple Correlation

- ▶ **Multiple Correlation:** It involves the study of three or more variables simultaneously. For example, if we study the relationship between the yield of wheat per acre and both amount of rainfall and the amount of fertilizers used, it becomes a problem relating to multiple correlation.

Linear and Non-linear Correlation

- ▶ **Linear Correlation:** The correlation between two variables is said to be **linear** if the amount of change in one variable tends to bear a constant ratio to the amount of change in other variable.
- ▶ **Non-linear (or Curvilinear):** The correlation between two variables is said to be **non-linear or curvilinear** if the amount of change in one variable does not bear a constant ratio to the amount of change in other variable.

Scatterplot: Bivariate points (x_i, y_i)

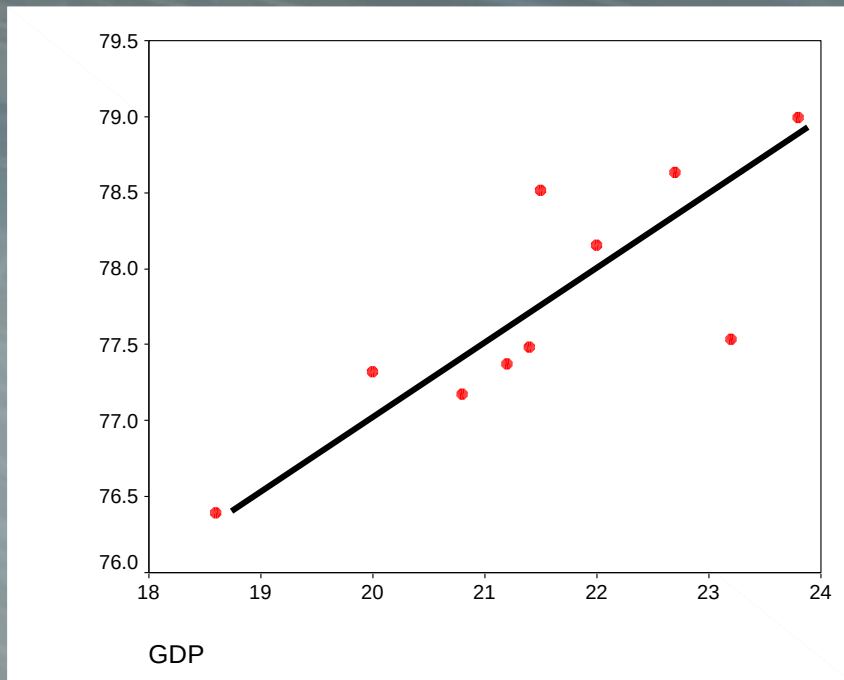


Interpreting Scatterplots

- ▶ **Form:** Can relationship be described by straight line (linear)? ..by a curved line? etc.
- ▶ **Outliers?:** Any deviations from overall pattern?
- ▶ **Direction** of the relationship either:
 - ▶ Positive association (upward slope)
 - ▶ Negative association (downward slope)
 - ▶ No association (flat)
- ▶ **Strength:** Extent to which points *adhere* to imaginary trend line

Example: Interpretation

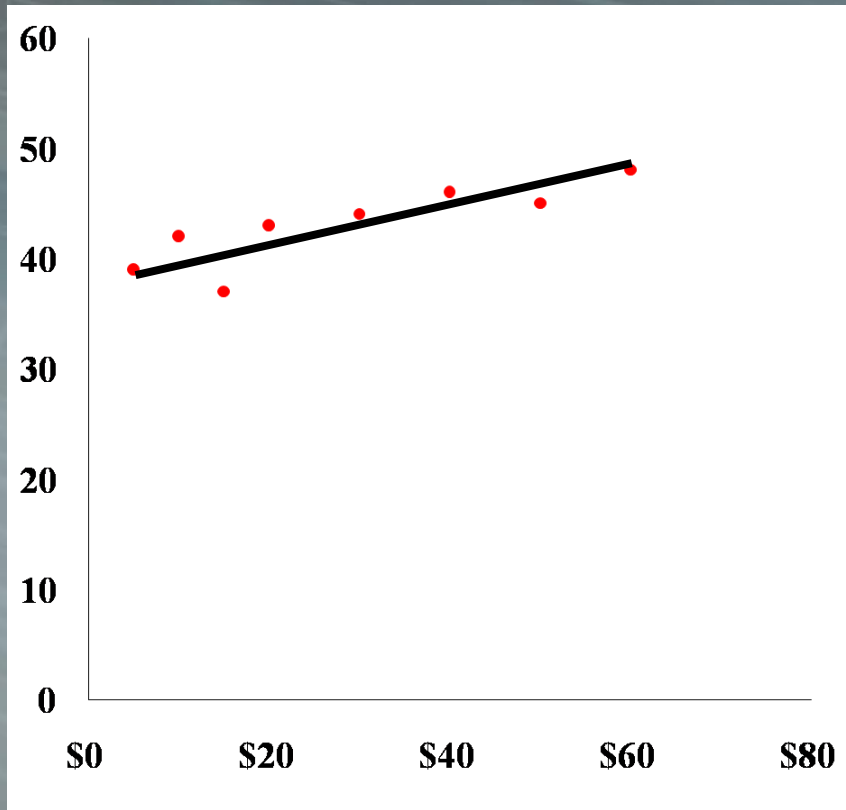
Here is the scatterplot we saw earlier:



Interpretation:

- ▶ Form: linear (straight)
- ▶ Outliers: none
- ▶ Direction: positive
- ▶ Strength: difficult to judge by eye

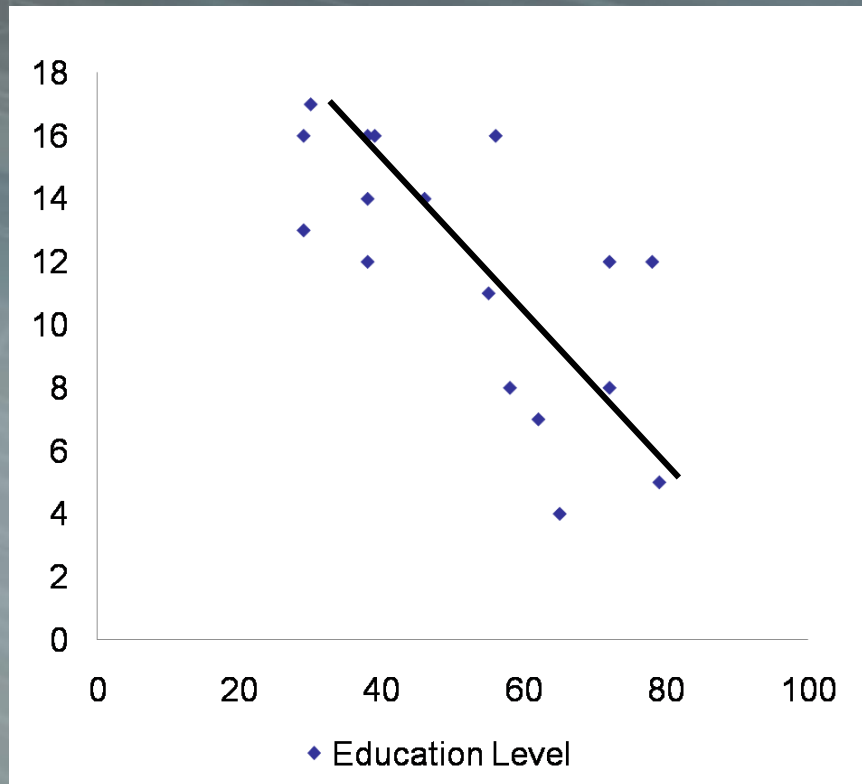
Example 2



Interpretation

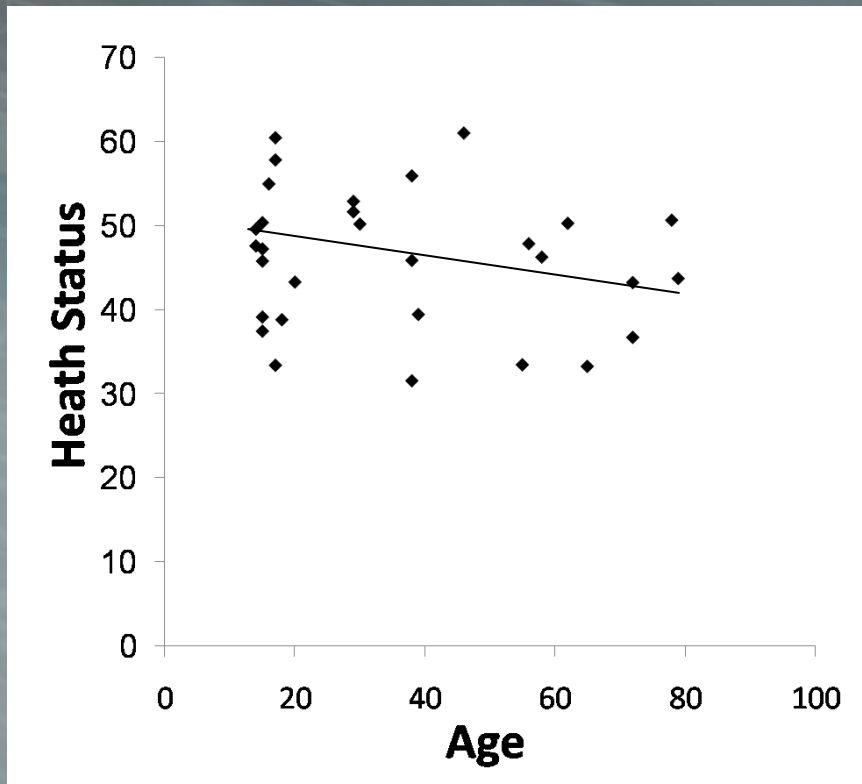
- ▶ Form: linear
- ▶ Outliers: none
- ▶ Direction: positive
- ▶ Strength: difficult to judge by eye (looks strong)

Example 3



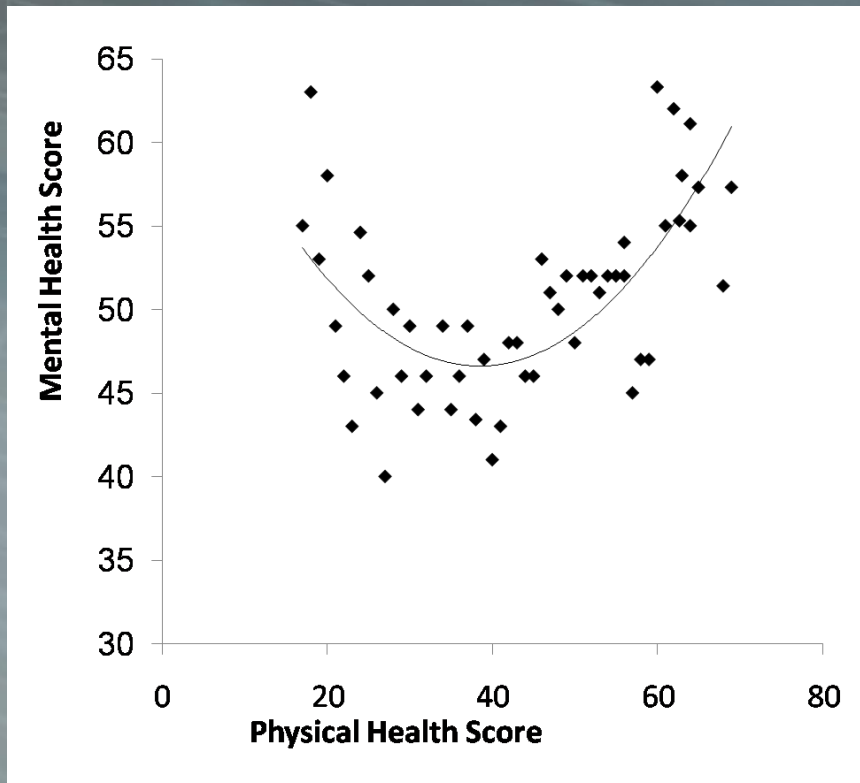
- ▶ Form: linear
- ▶ Outliers: none
- ▶ Direction: negative
- ▶ Strength: difficult to judge by eye (looks moderate)

Example 4



- ▶ Form: linear(?)
- ▶ Outliers: none
- ▶ Direction: negative
- ▶ Strength: difficult to judge by eye (looks weak)

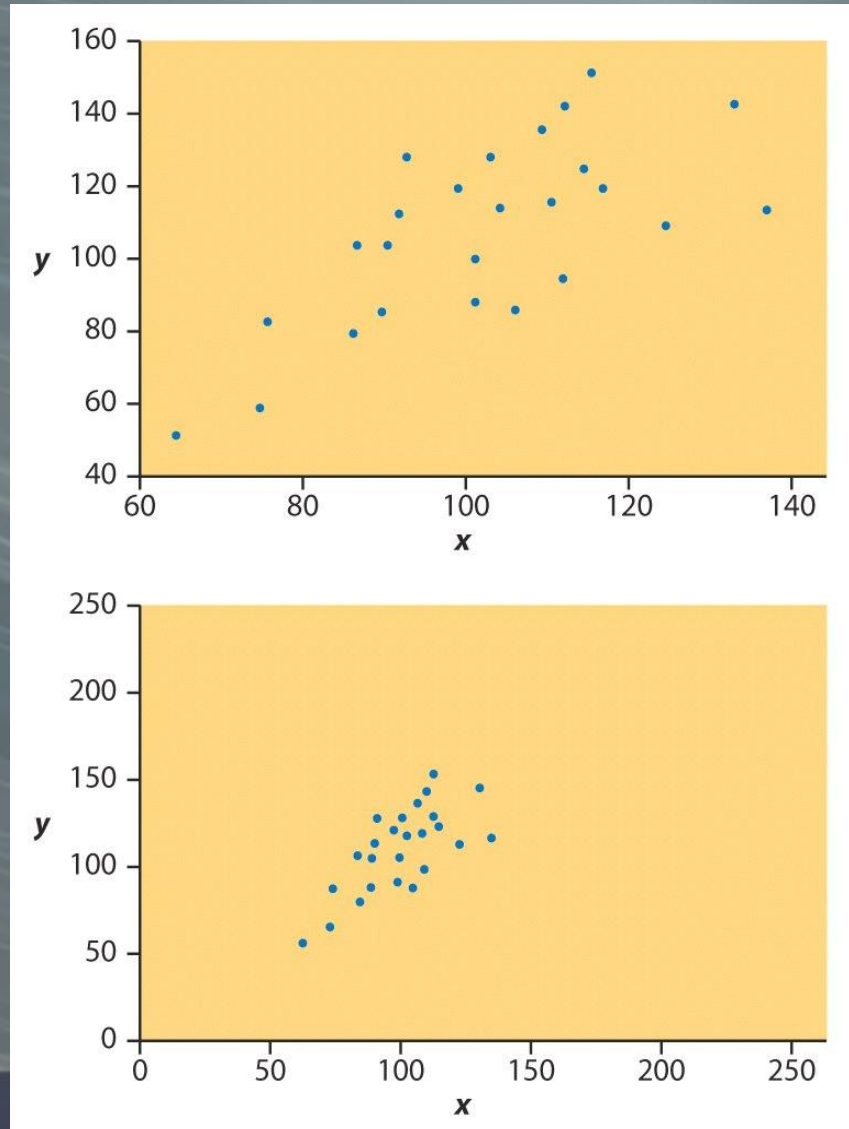
Interpreting Scatterplots



- ▶ Form: curved
- ▶ Outliers: none
- ▶ Direction: U-shaped
- ▶ Strength: difficult to judge by eye (looks moderate)

Correlational Strength

- ▶ It is difficult to judge correlational *strength* by eye alone
- ▶ Here are identical data plotted on differently axes
- ▶ First relationship *seems* weaker than second
- ▶ This is an artifact of the axis scaling
- ▶ We use a statistical called the *correlation coefficient* to judge strength objectively



Correlation coefficient (r)

- ▶ $r \equiv$ Pearson's **correlation coefficient**
- ▶ **Always between -1 and $+1$** (inclusive)
 - $r = +1 \Rightarrow$ all points on upward sloping line
 - $r = -1 \Rightarrow$ all points on downward line
 - $r = 0 \Rightarrow$ no line or horizontal line
- The closer r is to $+1$ or -1 , the **stronger** the correlation

Interpretation of r

- Direction: positive, negative, ≈ 0
- Strength: the closer $|r|$ is to 1, the stronger the correlation

$0.0 \leq |r| < 0.3 \Rightarrow$ weak correlation

$0.3 \leq |r| < 0.7 \Rightarrow$ moderate correlation

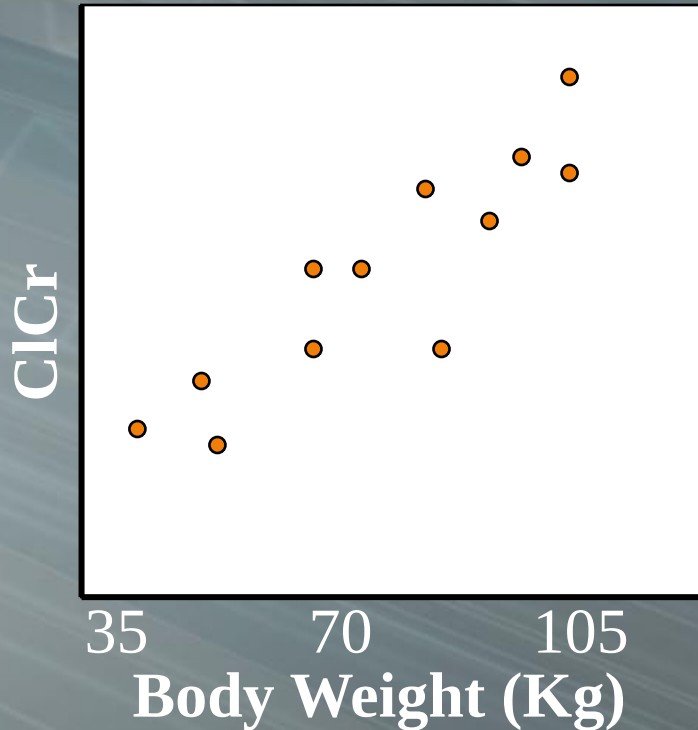
$0.7 \leq |r| < 1.0 \Rightarrow$ strong correlation

$|r| = 1.0 \Rightarrow$ perfect correlation

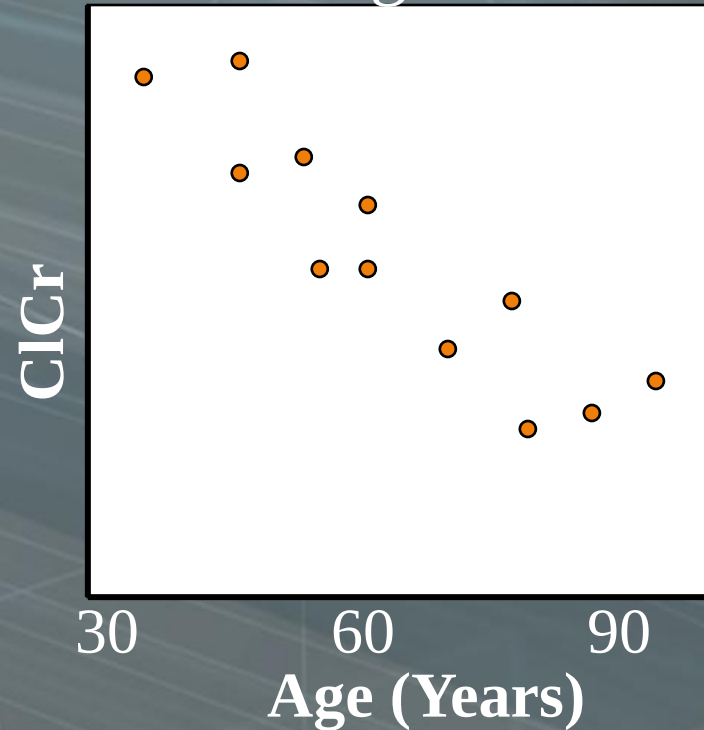
Linear Correlation

Creatinine clearance (ClCr)- measure of kidney function
Correlation is present in both cases

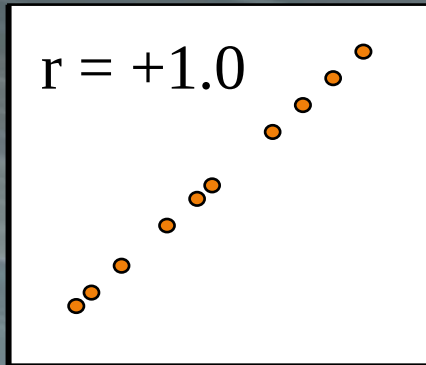
Positive



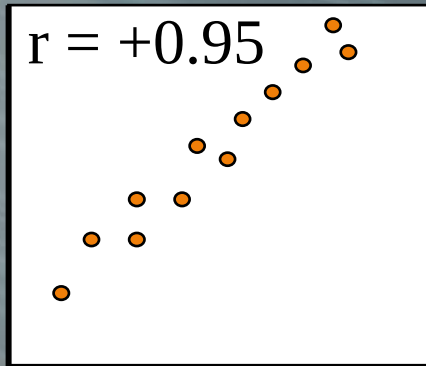
Negative



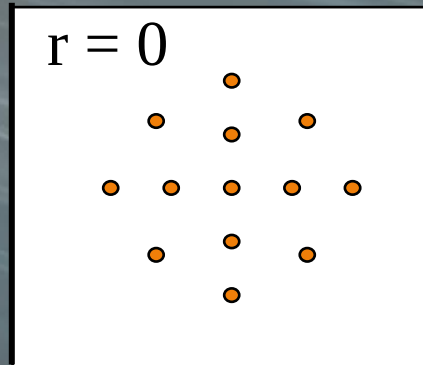
Linear Correlation



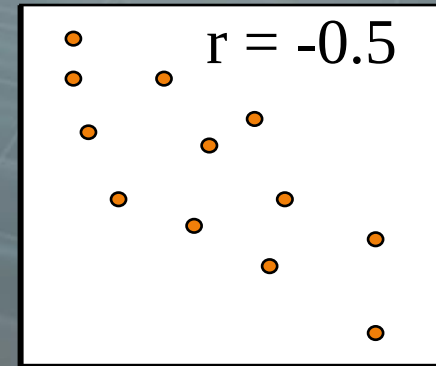
Perfect positive correlation



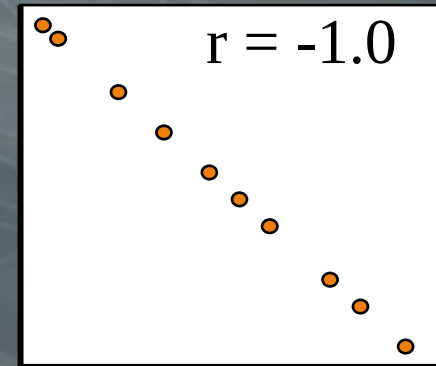
Partial positive correlation



No correlation



Partial negative correlation



Perfect negative correlation

Reminder: correlation
does not necessarily
imply causation
but.....

Calculating r

- ▶ Calculate mean and standard deviation of X
- ▶ Turn all X values into z scores
- ▶ Calculate mean and standard deviation of Y
- ▶ Turn all Y values into z scores
- ▶ Use formula on next page

Correlation coefficient r

$$r = \frac{1}{n-1} \sum_{i=1}^n z_X \cdot z_Y$$

where

$$z_X = \frac{x_i - \bar{x}}{s_x}$$

$$z_Y = \frac{y_i - \bar{y}}{s_y}$$

Example: Calculating r

X	Y	z_x	z_y	$z_x \cdot z_y$
21.4	77.48	-0.078	-0.345	0.027
23.2	77.53	1.097	-0.282	-0.309
20.0	77.32	-0.992	-0.546	0.542
22.7	78.63	0.770	1.102	0.849
20.8	77.17	-0.470	-0.735	0.345
18.6	76.39	-1.906	-1.716	3.271
21.5	78.51	-0.013	0.951	-0.012
22.0	78.15	0.313	0.498	0.156
23.8	78.99	1.489	1.555	2.315
21.2	77.37	-0.209	-0.483	0.101
			7.285	

Mean of $x = 21.52$; $stddev$ of $x = 1.532$;
Mean of $y = 77.754$; $stddev$ of $y = 0.795$

Example: Calculating r

$$\begin{aligned} r &= \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) \\ &= \left(\frac{1}{10-1} \right) (7.285) \\ &= 0.809 \end{aligned}$$

$r = .81 \Rightarrow$ strong positive correlation

Linear Regression

What is Regression?

- ▶ Correlation did a good job in qualifying the relationship between variables
- ▶ But, is there more precise way to represent relationships?
 - ▶ A mathematical model !!
 - ▶ Like many physical laws , e.g. $F=ma$, to represent relation with function

$$y = F(x)$$

What is Regression?

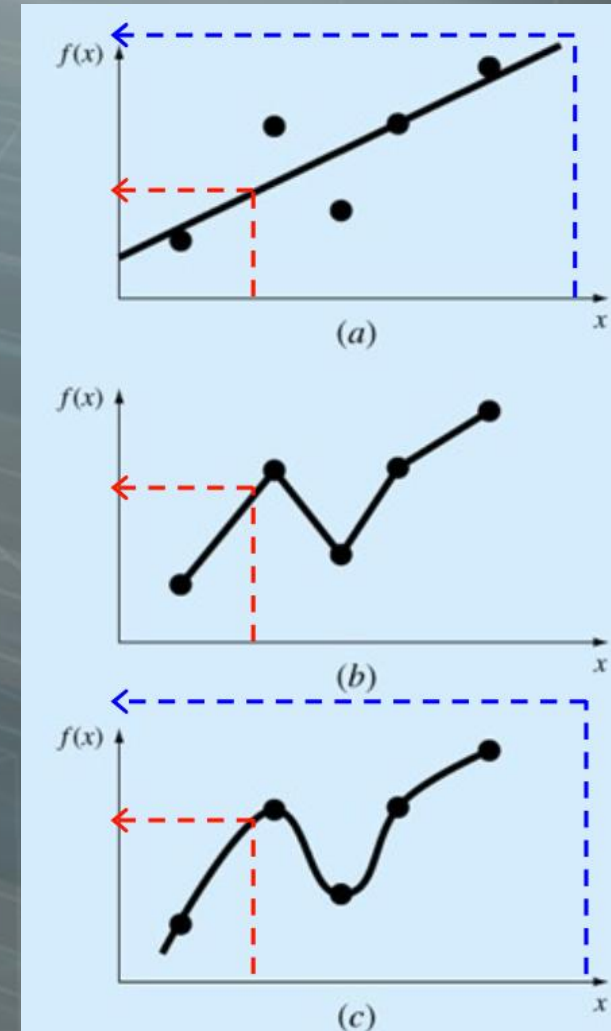
- ▶ Function: a mathematical relationship enabling us to predict what values of one variable (y) correspond to given values of another variable (x).

$$y = F(x)$$

- ▶ y: is referred to as the **dependent variable**, the response variable or the predicted variable.
- ▶ x: is referred to as the **independent variable**, the explanatory variable or the predictor variable.

What is Regression?

- ▶ Use the data to “**fit**” into the model (i.e. function)
- ▶ **Fitting** : find the parameters in the model
 - ▶ Depends on degree of freedom in your model, the number of parameters is different!
 - ▶ E.g. linear function : 2 parameters



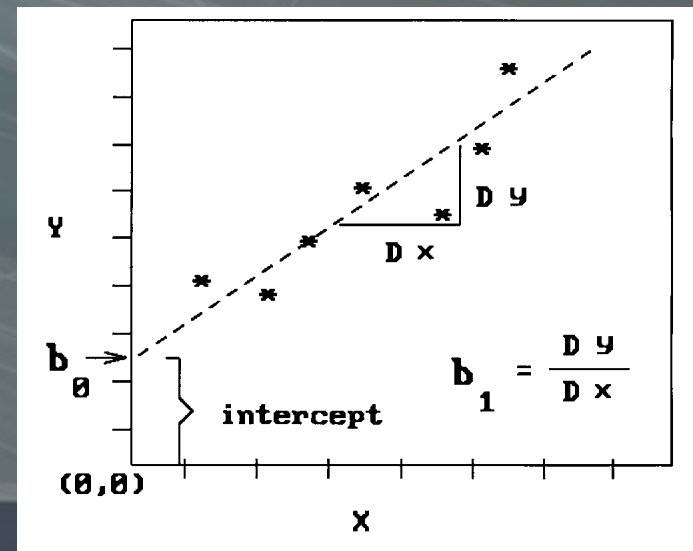
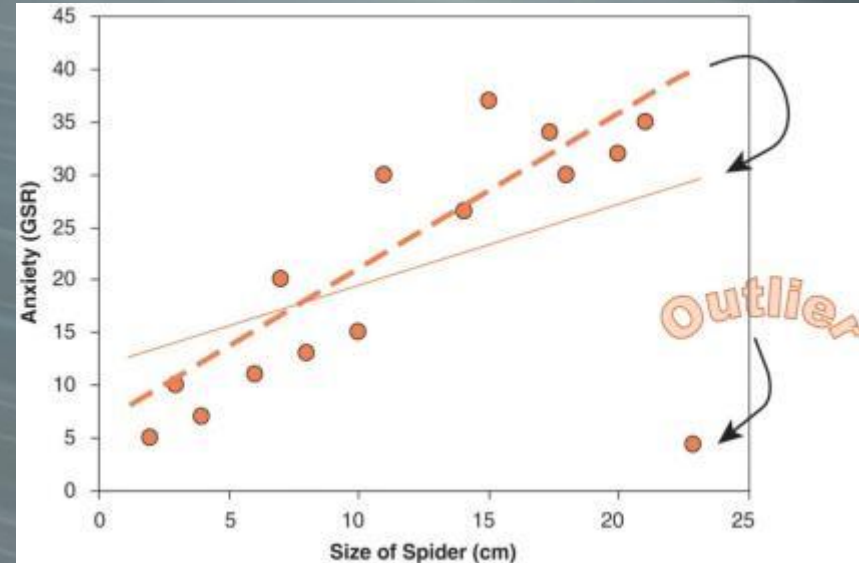
Simple Linear Regression

- ▶ One of the simplest function is a straight line

$$y = b_0 + b_1x$$

- ▶ b_0 is the intercept,
- ▶ b_1 is the slope.

- ▶ The regression is to find out both b_0 and b_1



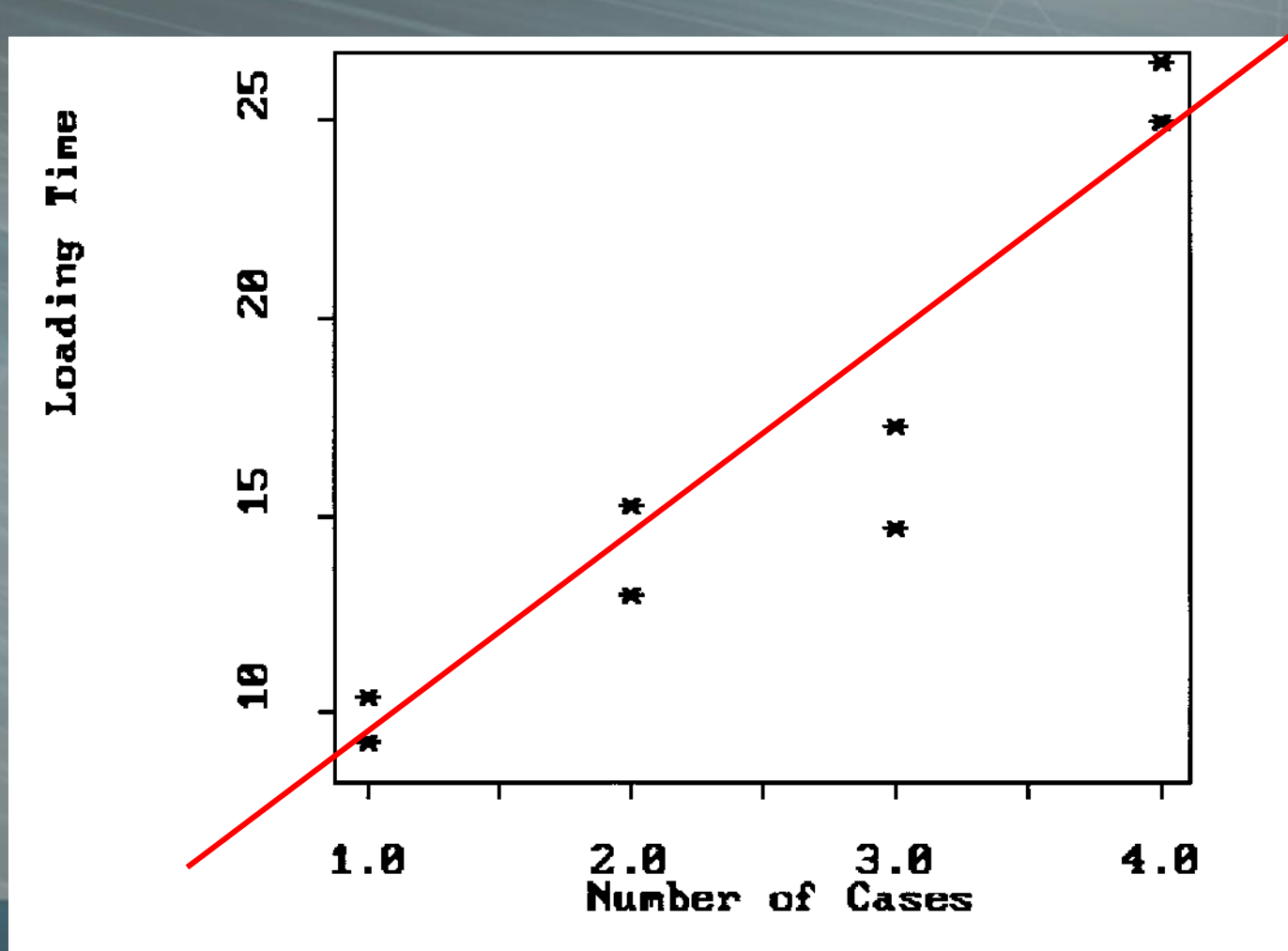
Examples

- ▶ The time needed to fill a soft drink vending machine  ▶ The number of cases needed to fill the machine
- ▶ The mean piglet weight  ▶ The piglet size
- ▶ Maintenance cost of tractors  ▶ The age of the tractor

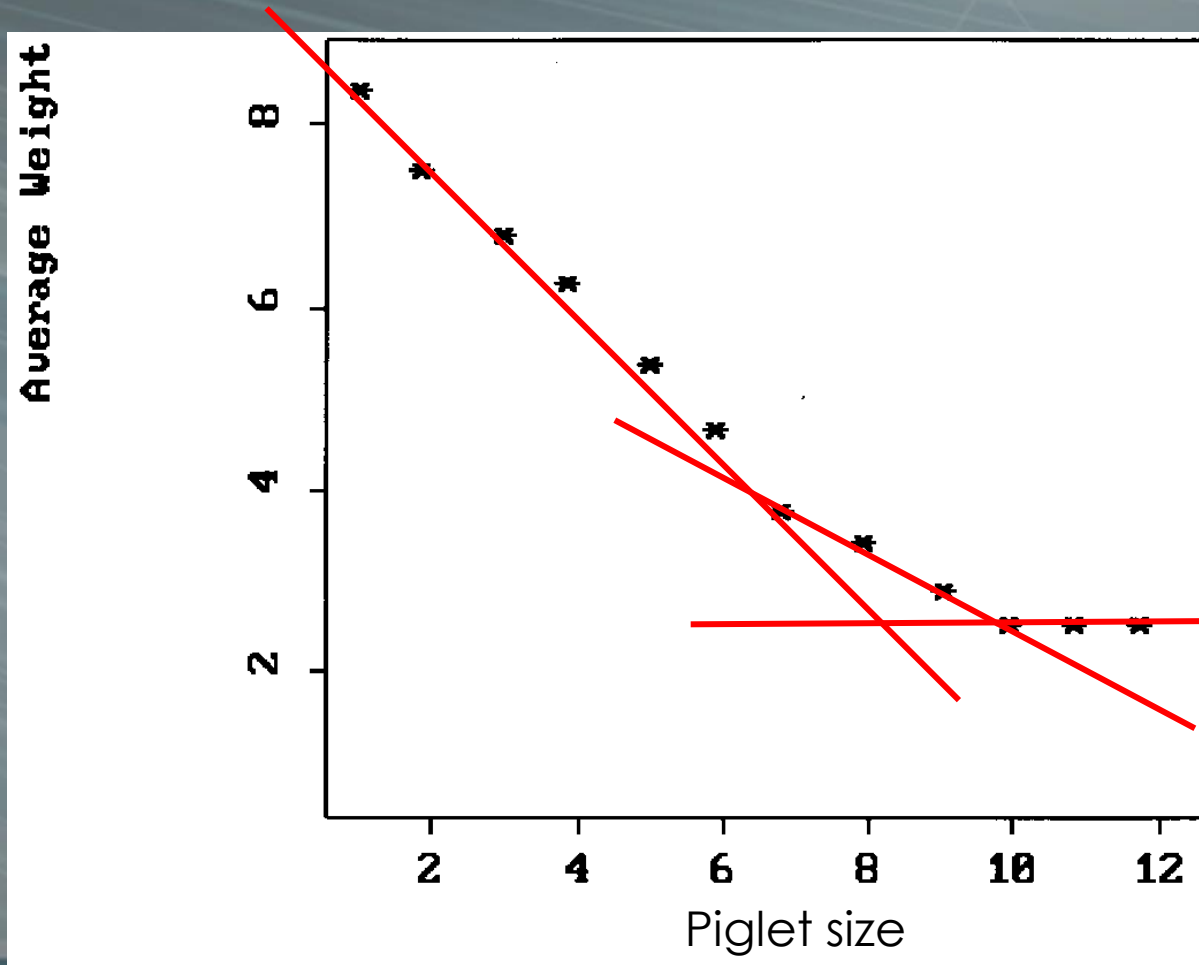
In each case, the statement can be read as; ***y is a function of x.***

Usually we take the variable which we can control more easily as the independent variables, i.e. x

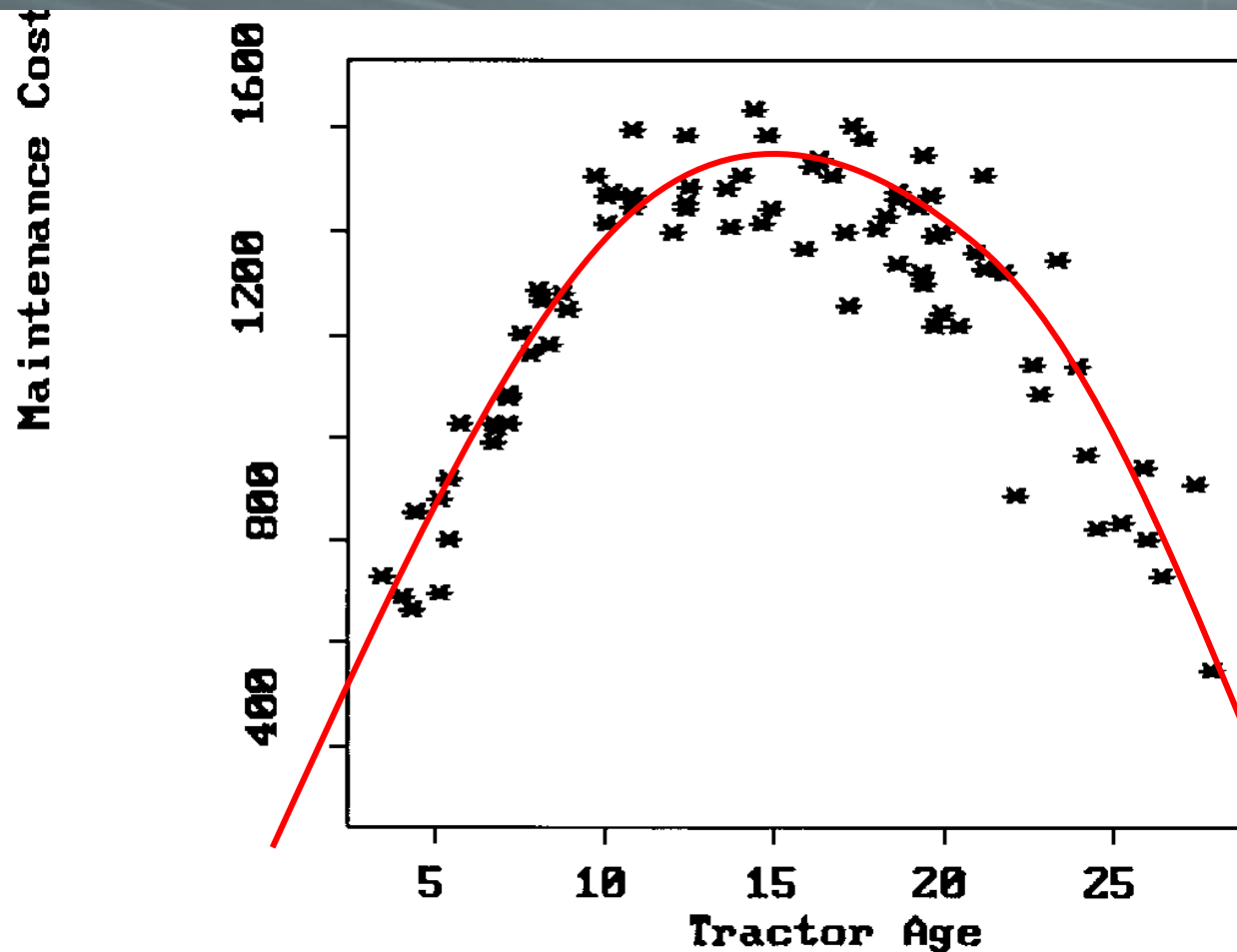
An operations supervisor measured how long it takes one of her drivers to put 1, 2, 3 and 4 cases of soft drink into a soft drink machine. In this case the levels of the explanatory variable, x are $\{1,2,3,4\}$, and she controls them. She might repeat the measurement a couple of times at each level of x . A scatter plot of the resulting data might look like:



A farm manager is interested in the relationship between piglet size and average piglet weight. She examines the farm records over the last couple of years and records the piglet size and average weight for all births. A plot of the data pairs looks like the following:



A farm operations student is interested in the relationship between maintenance cost and age of farm tractors. He performs a telephone interview survey of the 52 commercial potato growers in the County. One part of the questionnaire provides information on tractor age and maintenance cost (fuel, lubricants, repairs, etc). A plot of these data might look like:



Linear Regression: Line fitting

- ▶ For a particular data i

$$y_i = b_0 + b_1 x_i + e_i$$

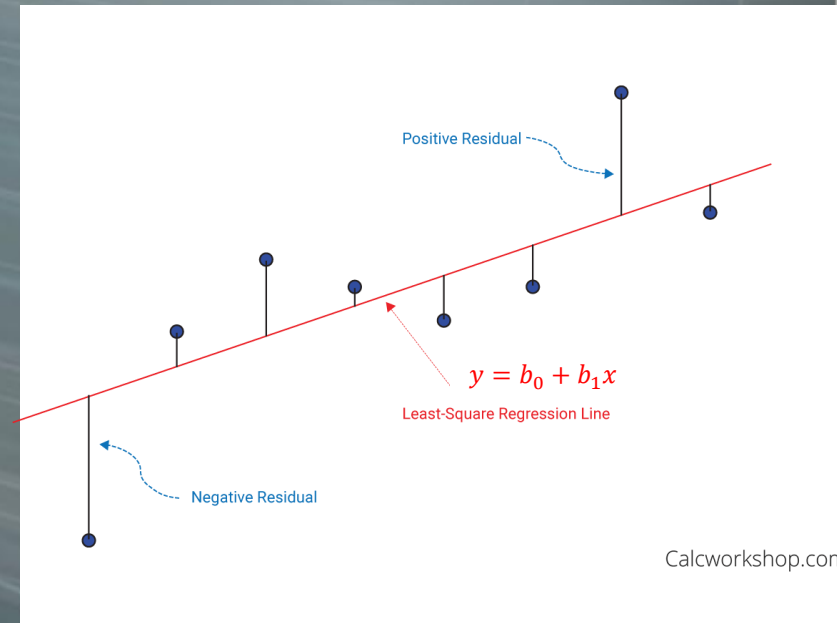
Straight line

Random error
(residuals)

- ▶ Distance of points from line = residuals e_i
- ▶ **Minimizes** sum of square residuals e_i

$$E(b_0, b_1) = \min_{b_0, b_1} \sum_{i=1}^n \varepsilon_i^2$$

- ▶ Least squares regression line



Simplest assumption: $e_i \sim N(0, \sigma^2)$ for all i , and e_i 's are independent

Least Squares Solution

- ▶ Want to find “best” $\mathbf{b}_0$ and $\mathbf{b}_1$
- ▶ We minimize $\mathbf{E}(\mathbf{b}_0, \mathbf{b}_1)$
 - $= \sum e_i^2$
 - $= \sum (\mathbf{y}_i - (\mathbf{b}_0 + \mathbf{b}_1 \mathbf{x}_i))^2$
- ▶ Use calculus:
 - ▶ take derivative with respect to $\mathbf{b}_0$ and with respect to $\mathbf{b}_1$,
 - ▶ then set the two resulting equations equal to zero and solve for $\mathbf{b}_0$ and $\mathbf{b}_1$

$$\frac{\partial E}{\partial \beta_0} = 0$$

$$\frac{\partial E}{\partial \beta_1} = 0$$

Least Squares Solution

- ▶ We skip the derivation, and the solution:

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

S_{xy} : Sums of cross products of x and y.

S_{xx} : Sums of squares of x.

Sums of Squares

$$\begin{aligned}S_{xx} &= \sum_{i=1}^n (x_i - \bar{x})^2 \\&= (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2 \\&= \sum_{i=1}^n (x_i^2) - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2\end{aligned}$$

Sums of
squares of
x.

$$\begin{aligned}S_{xy} &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\&= (x_1 - \bar{x})(y_1 - \bar{y}) + \cdots + (x_n - \bar{x})(y_n - \bar{y}) \\&= \sum_{i=1}^n (x_i y_i) - \frac{1}{n} \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)\end{aligned}$$

Sums of
cross
products
of x and
y.

Example

A company that repairs small computers needs to develop a better way of providing customers typical repair cost estimates. To begin this process, they compiled data on repair times (in minutes) and the number of components needing repair or replacement from the previous week. The data, sorted by number of components are as follows:

	Number of components	Repair time
i	x_i	y_i
1	1	23
2	2	29
3	4	64
4	4	72
5	4	80
6	5	87
7	6	96
8	6	105
9	8	127
10	8	119
11	9	145
12	9	149
13	10	165
14	10	154

Example: Parameter estimates

$$b_1 = \frac{s_{xy}}{s_{xx}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$
$$b_0 = \bar{y} - b_1 \bar{x}$$

Easy to compute using a software program with a statistical analysis package.

$$b_0 = 7.71$$

$$b_1 = 15.20$$

The regression result :

$$y = 7.71 + 15.20x$$

Steps in a Regression Analysis

1. Examine the scatterplot of the data.
 - ▶ Does the relationship look linear?
 - ▶ Are there points in locations they shouldn't be? (i.e. outliers)
 - ▶ Do we need a transformation?
2. Assuming a linear function looks appropriate, estimate the regression parameters.
 - ▶ How do we do this? (Method of Least Squares)
3. Do a validation of your least square regression line
 - ▶ Examine the residuals. Is there evidence that a more complicated relationship (say a polynomial) should be considered

Validating Linear Regression

- ▶ What is validation and why do it?
 - ▶ We need to know how well our models will perform
 - ▶ Do a split between training data and test data
 - ▶ Model built on training data
 - ▶ Test/validation data then passed through model
 - ▶ We should never test a model on the same data that was used to build it – this would be confirming a hypothesis with the same hypothesis
 - ▶ Try model in the real world

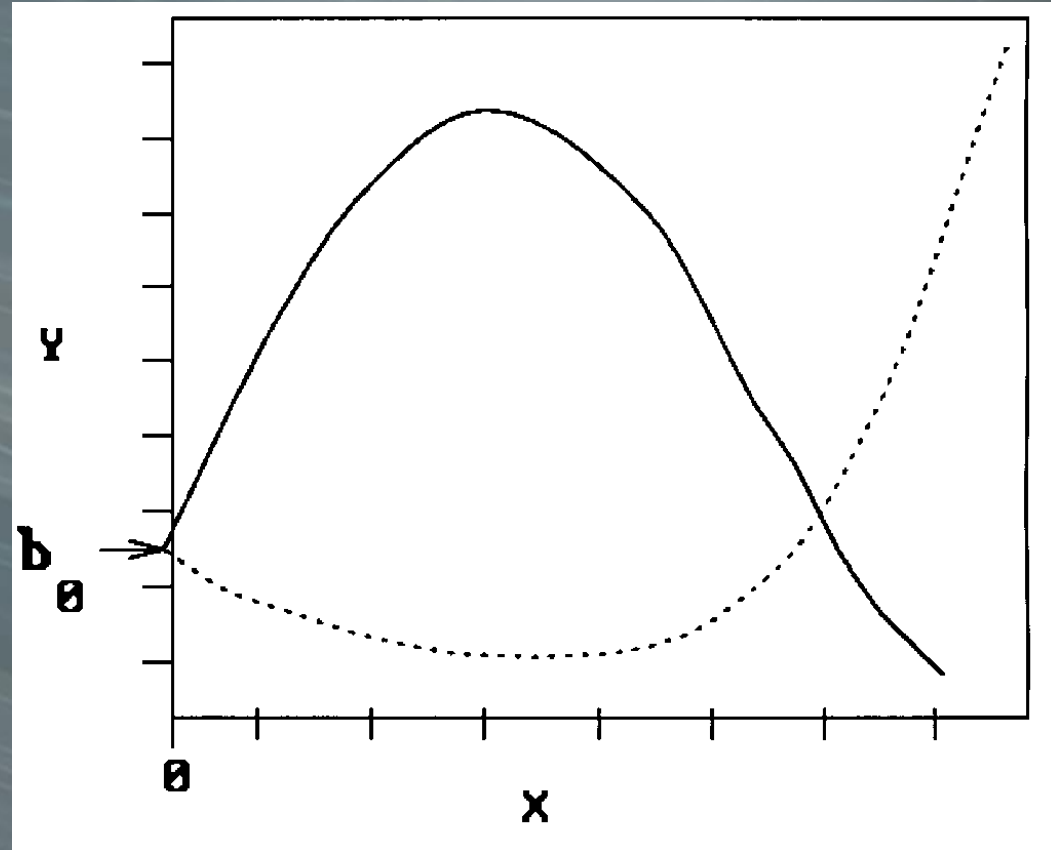
Curve Fitting

- ▶ Is a more general term than regression
- ▶ It refers to the process of fitting different types of curve through sets of data
- ▶ Typically involves:
 - ▶ Polynomials
 - Nonlinear:
 - ▶ Sinusoidal curves
 - ▶ Exponential curves

Example: Polynomial relationship

$$y = b_0 + b_1x + b_2x^2$$

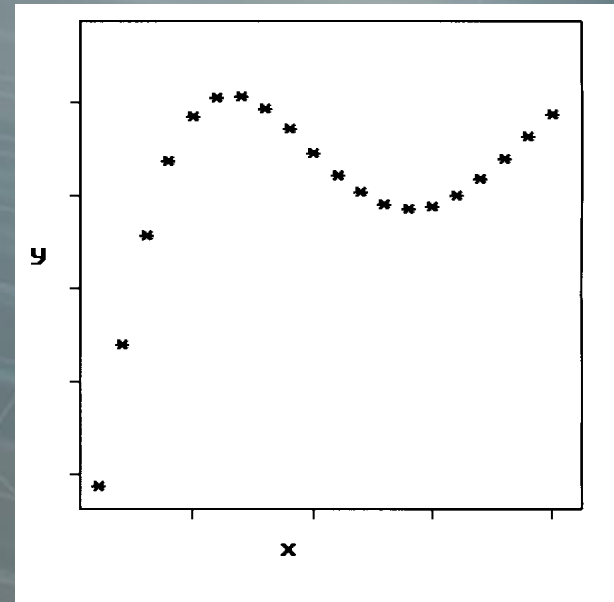
b_0 : intercept,
 b_1 : linear coefficient,
 b_2 : quadratic coefficient.



Nonlinear relationship:

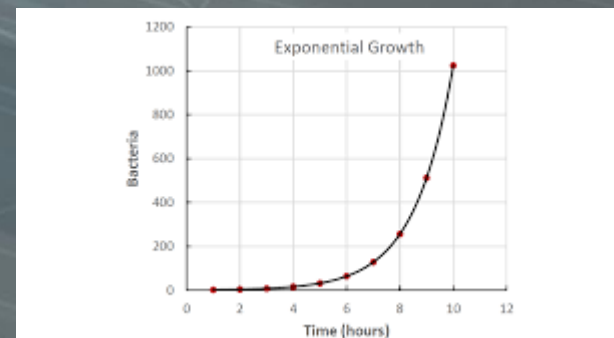
Sinusoidal curves

$$y = b_0 \sin(b_1 x + b_2 x^2)$$



Exponential curves

$$y = b_0 b_1^x$$

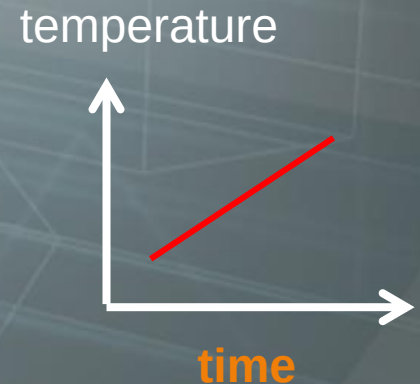


Marking Decision

- ▶ By knowing the relationship of data, we can **make better decision**
- ▶ Using the example of temperature and electricity consumption
- ▶ The electricity provider can base on the temperature to adjust the generation of electricity needed
 - ▶ Better utilize the resources, maximize profit

Predicting the Future

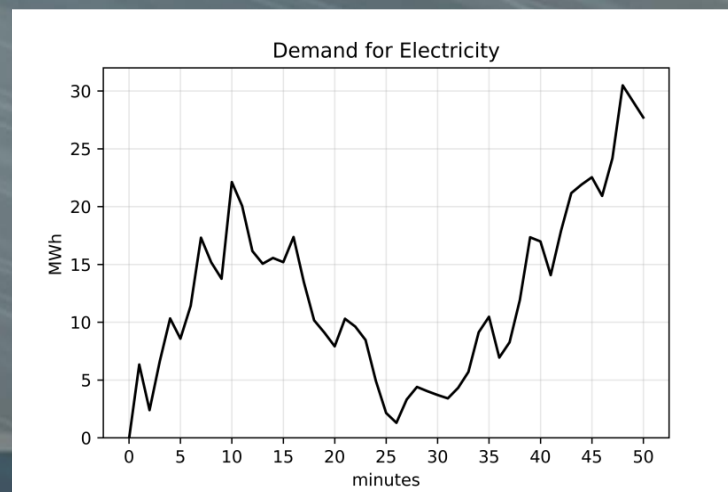
- ▶ Linear regression can also be used to predict / plan for the future
- ▶ It is the same as finding a **variable's relationship with time**
- ▶ If we have one of the two variables as time, usually x will become time, we can use regression to understand their underlying relationship



Time Series Forecasting

Time Series

- ▶ Time series data is common across many industries.
- ▶ Finance: stock prices, asset prices, macroeconomic factors
- ▶ E Commerce: page views, new users, searches
- ▶ Business: transactions, revenue, inventory levels



Forecasting

- ▶ Then, we are going to find the effect of time to a particular data/parameter
- ▶ This also refers to “Forecasting”
- ▶ There are many methods to make reliable forecasts
- ▶ These might be long, medium or short term
- ▶ But there is no single best way of forecasting

The Aim of Forecasting

- ▶ Is to make the best possible predictions for future events, minimise errors, and provide the most reliable information possible
- ▶ It provided valuable information for management decisions
- ▶ It does not (necessarily) use the most sophisticated and obscure mathematics

Example

- ▶ We have collected the **price** of a certain raw material (e.g. oil, timber, etc) as below:

Month	1	2	3	4	5	6
Price	5.3	6.4	7.5	5.6	9.7	10.2

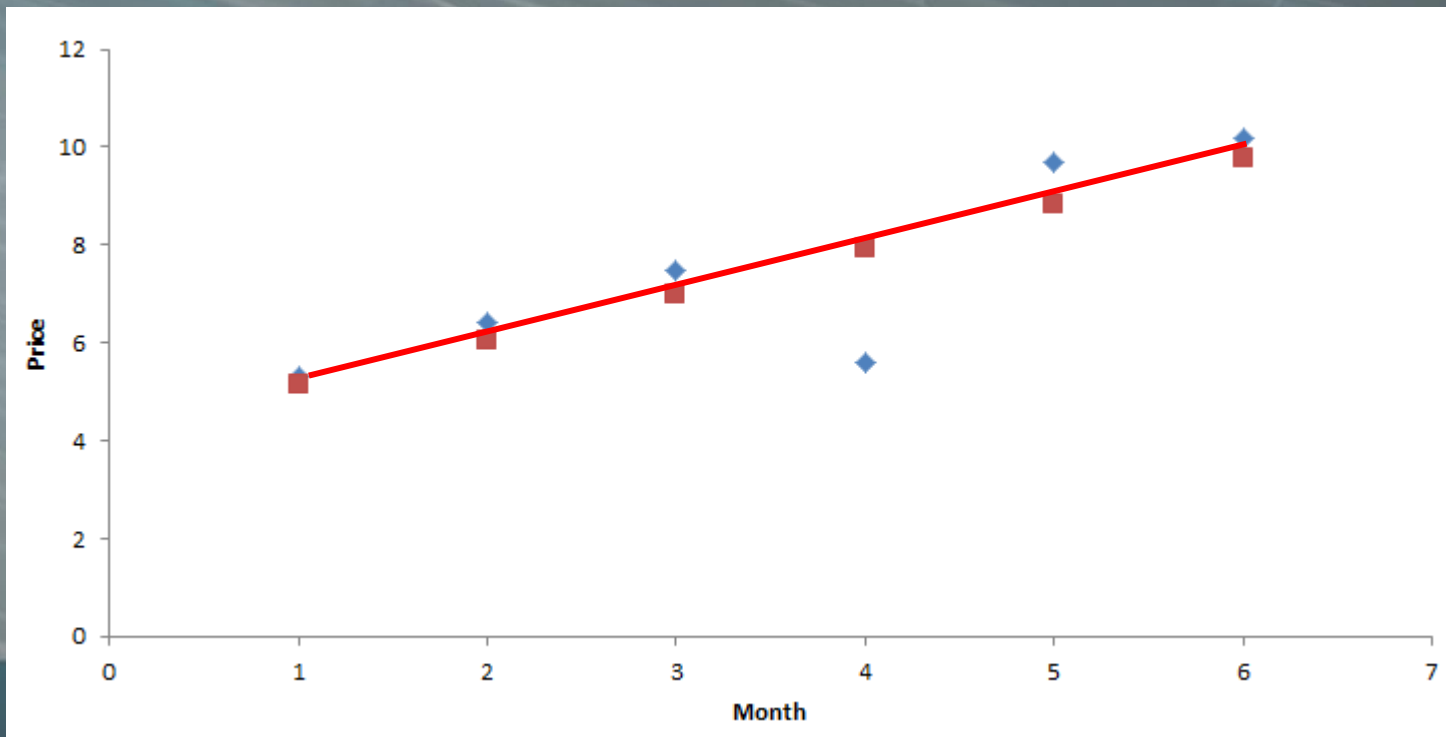
- ▶ Our objective is to understand the trend of the price, and possibly to **forecast what the price will be in month 7**
- ▶ One first approach is to plot it in a graph

Forecasting the Future

- ▶ But it is not enough, as how we predict the price in month 7?
- ▶ Remember, linear regression helps us to know the **underlying relationship** of the parameters
- ▶ So, if we know the line of best fit, we can **extend it to 7th month!!** And we can do a forecast

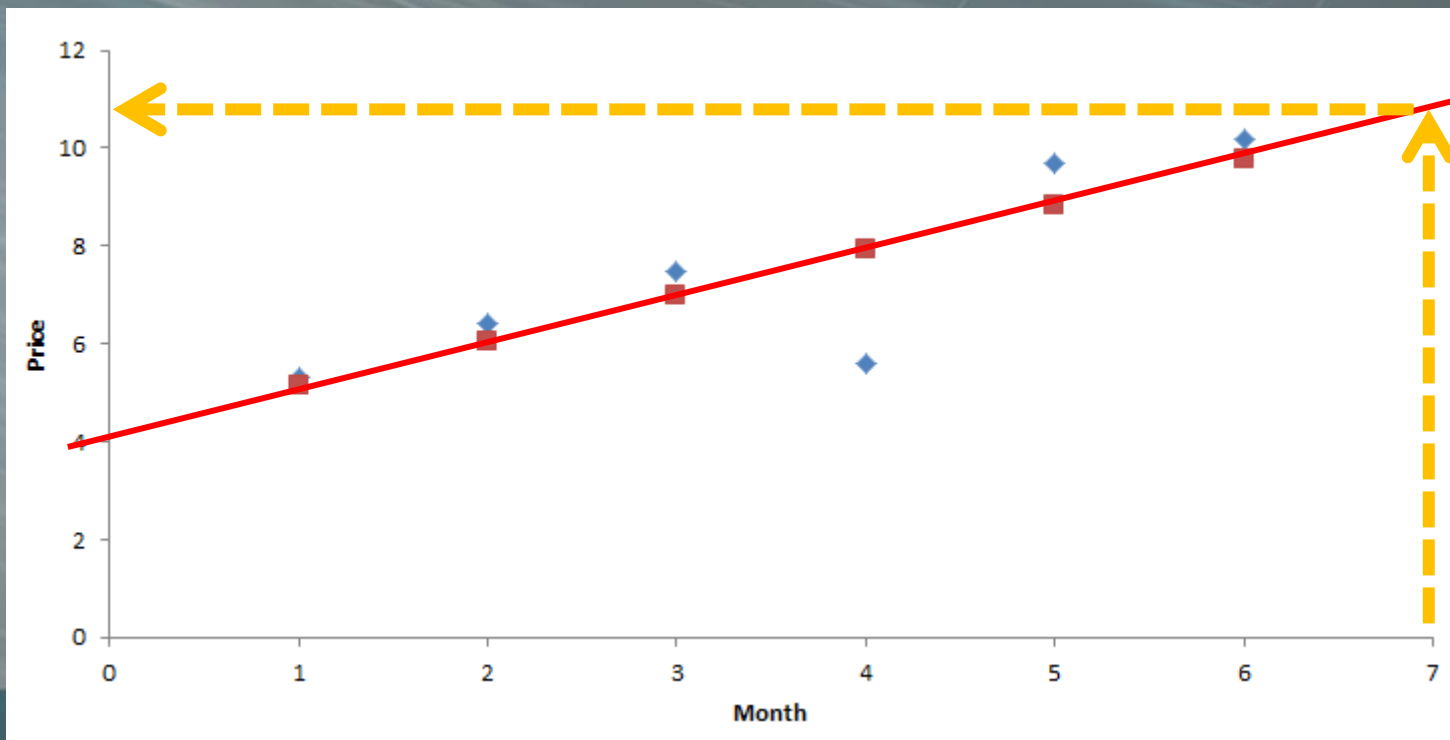
Forecasting with linear regression

- ▶ Performing linear regression, we have the fitted line which represents our estimated underlying relationship of the price and time
- ▶ So it does not limited to the data range



Forecasting the Future

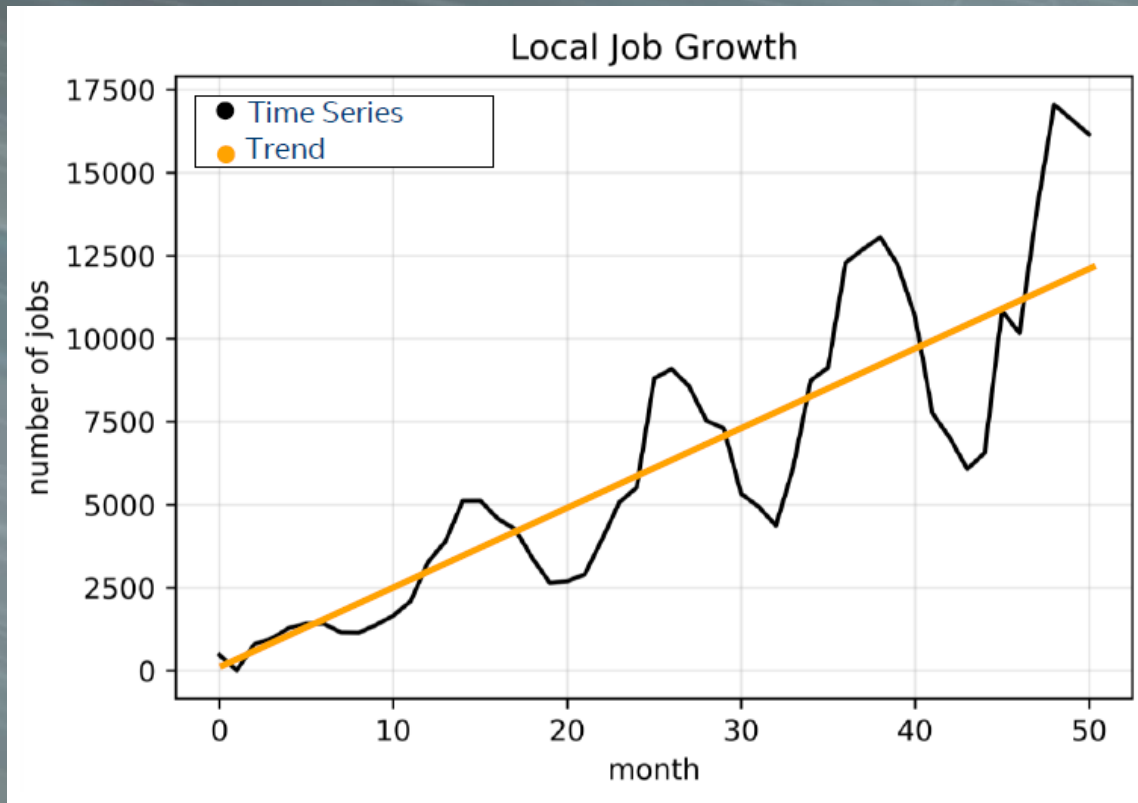
- ▶ As a result, we can estimate the price at 7th month using the line
- ▶ The estimated price should be around 10.7



Time Series Components

- ▶ A time series has three components
 - ▶ Trend : long term direction
 - ▶ Seasonality : periodic behavior
 - ▶ Residual : irregular fluctuations

Trend

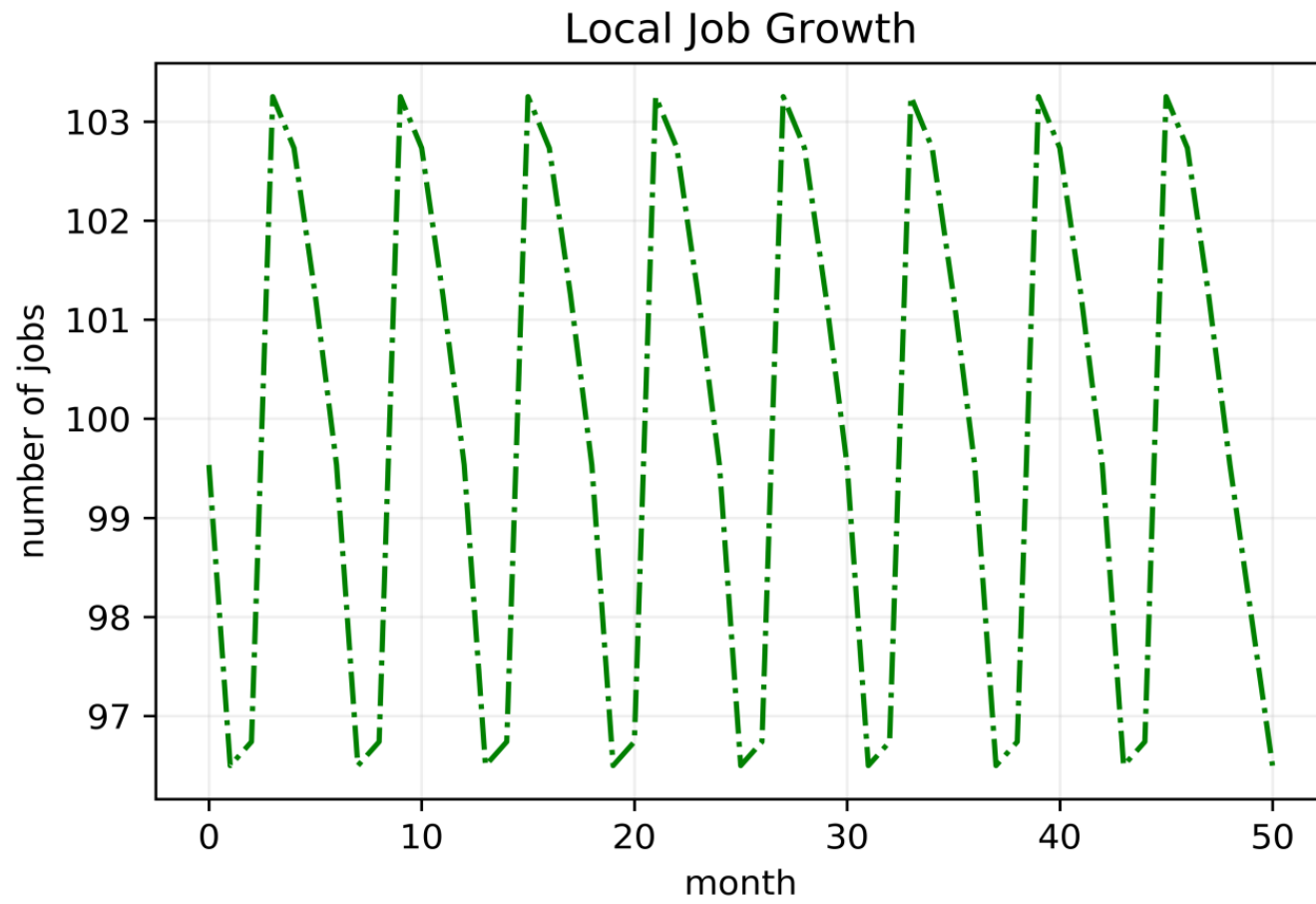


Seasonality

Seasonality captures effects that occur with specific frequency. It can be driven by many factors.

- ▶ Naturally occurring events such as weather fluctuations caused by time of year
- ▶ Business or administrative procedures , such as start and end of a school year
- ▶ Social or cultural behavior , such as holidays or religious observances
- ▶ Fluctuations due to calendar events , such as the number of Mondays per month for trading or holidays that shift from year to year Easter Chinese New Year

Seasonality

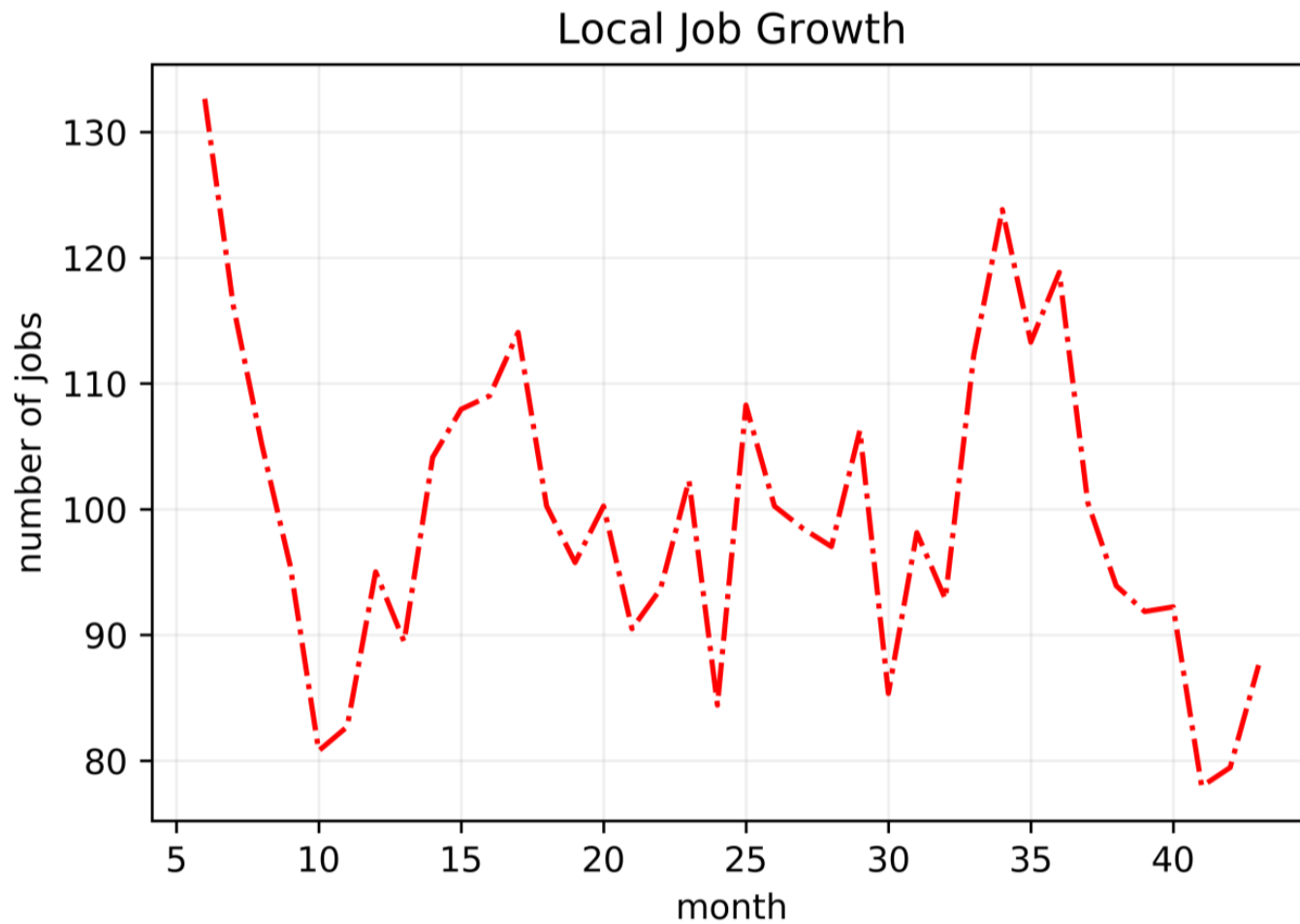


Residuals

Residuals are the random fluctuations left over after trend and seasonality are removed.

- ▶ They are what is left over after trend and seasonality are removed from the original time series
 - ▶ You should not see a trend or seasonal pattern in the residual
- ▶ They represent short term fluctuations
- ▶ They're either random or a portion of the trend or seasonality components was missed in the decomposition

Residuals



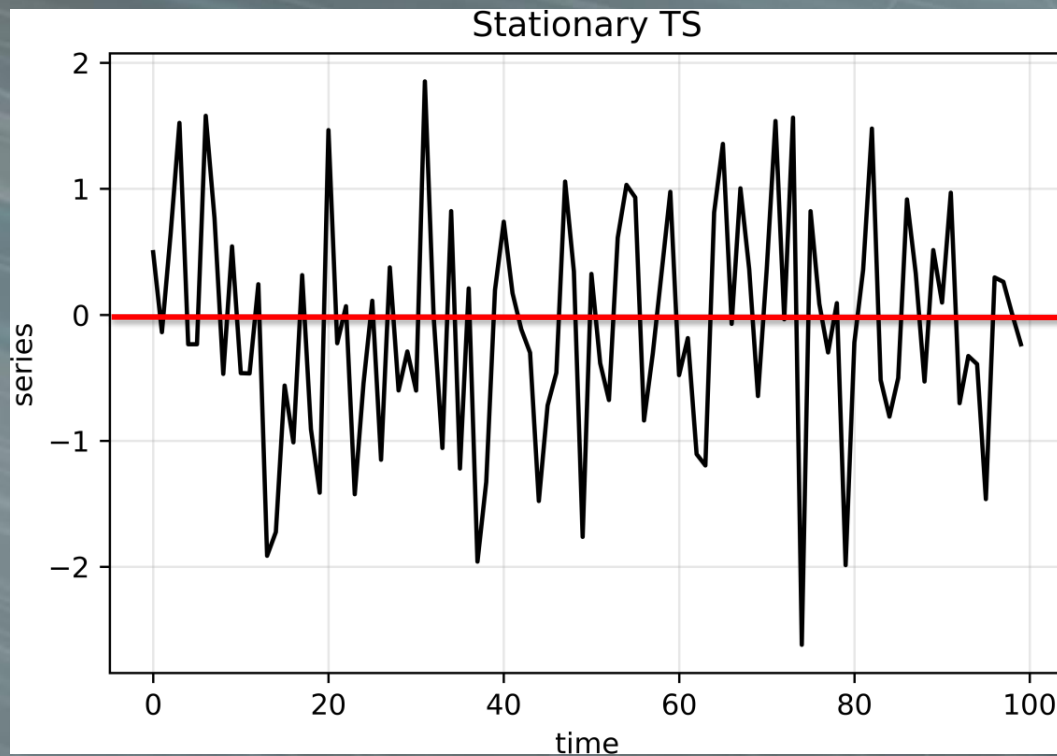
Stationarity

Source: Intel Time Series 501

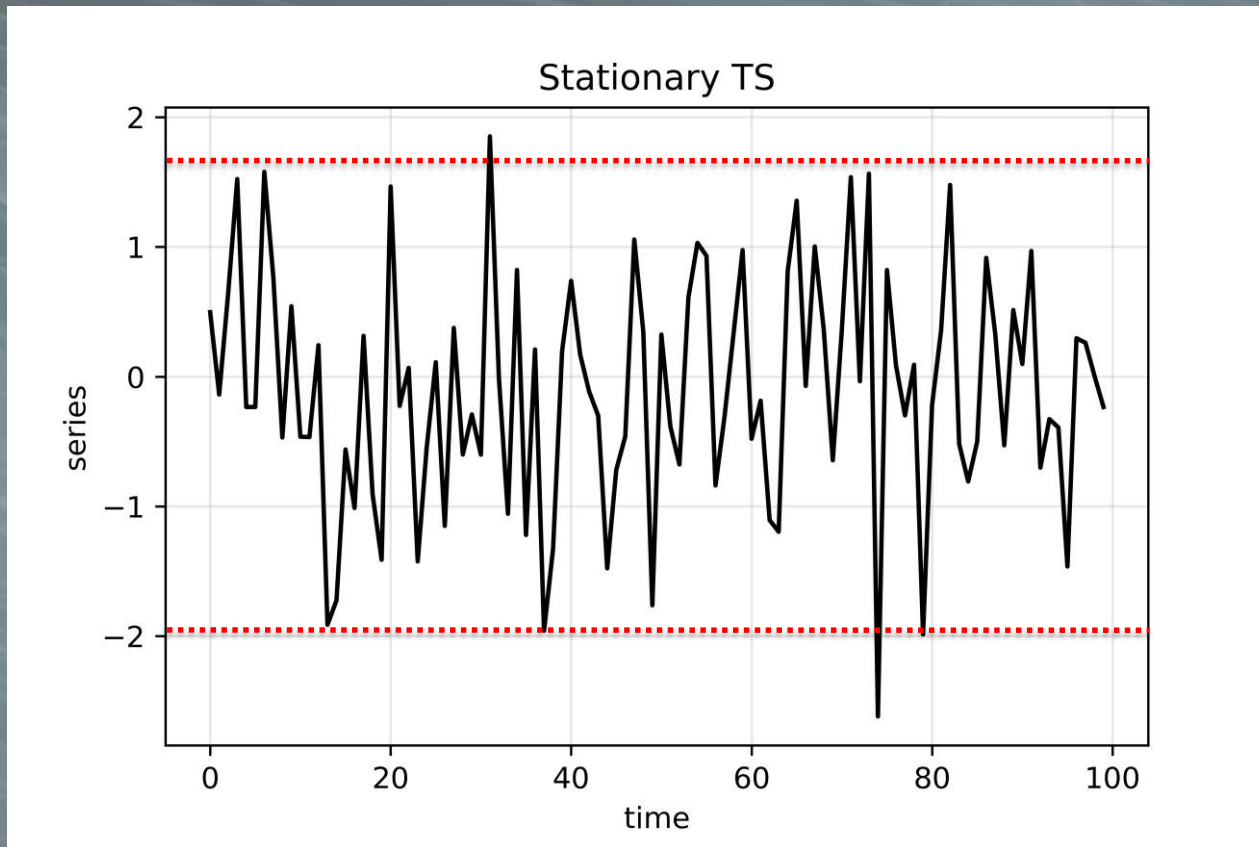
What Is Stationarity?

- ▶ A stationary time series is one whose properties do not depend on the time at which the series is observed.
- ▶ There are no changes in the underlying system.
 - ▶ Constant mean (no trend)
 - ▶ Constant variance (no heteroscedasticity)
 - ▶ Constant autocorrelation structure
 - ▶ No periodic component (no seasonality)

Assumption 1: Constant Mean



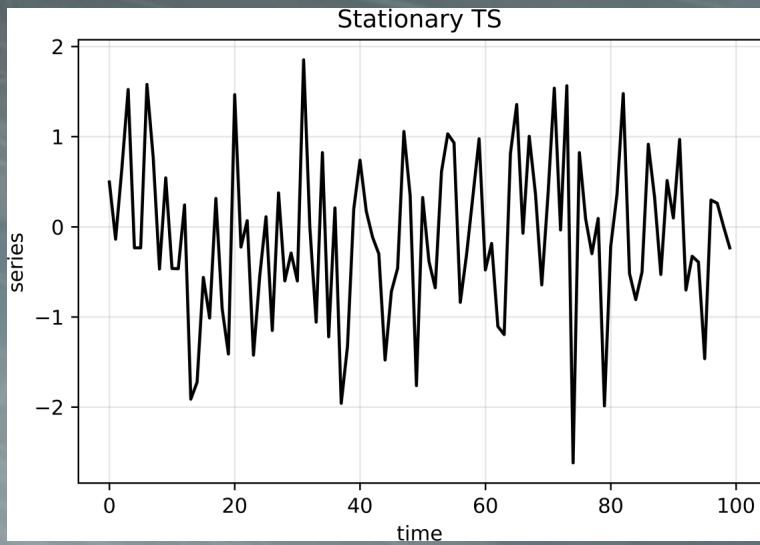
Assumption 2: Constant Variance



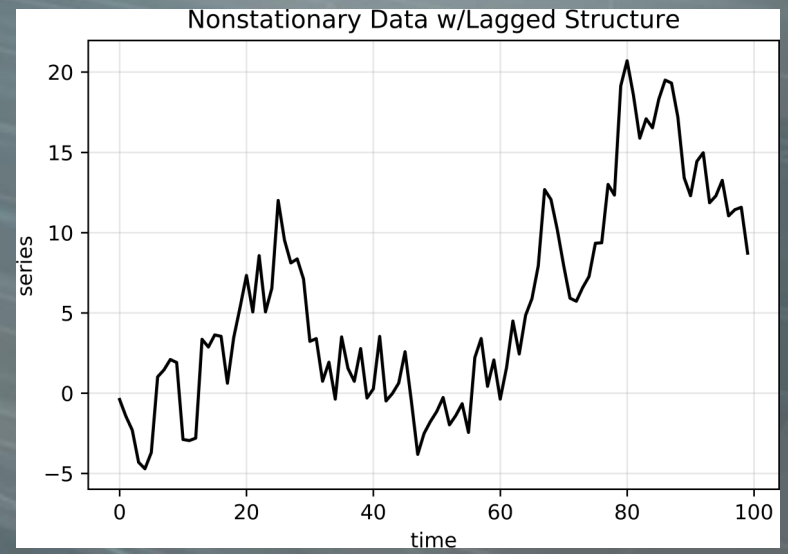
Autocorrelation

Autocorrelation is a key concept in time-series analysis.

- ▶ Autocorrelation is the correlation between a measurement at two different times.
- ▶ The time interval between values is called the lag.
- ▶ For example, stock prices may be correlated from one day to the next with a lag value of 1.
- ▶ Autocorrelation often results in a pattern, whereas a time series without autocorrelation will exhibit randomness.



No autocorrelation



Has autocorrelation

Assumption 3: Constant Autocorrelation Structure

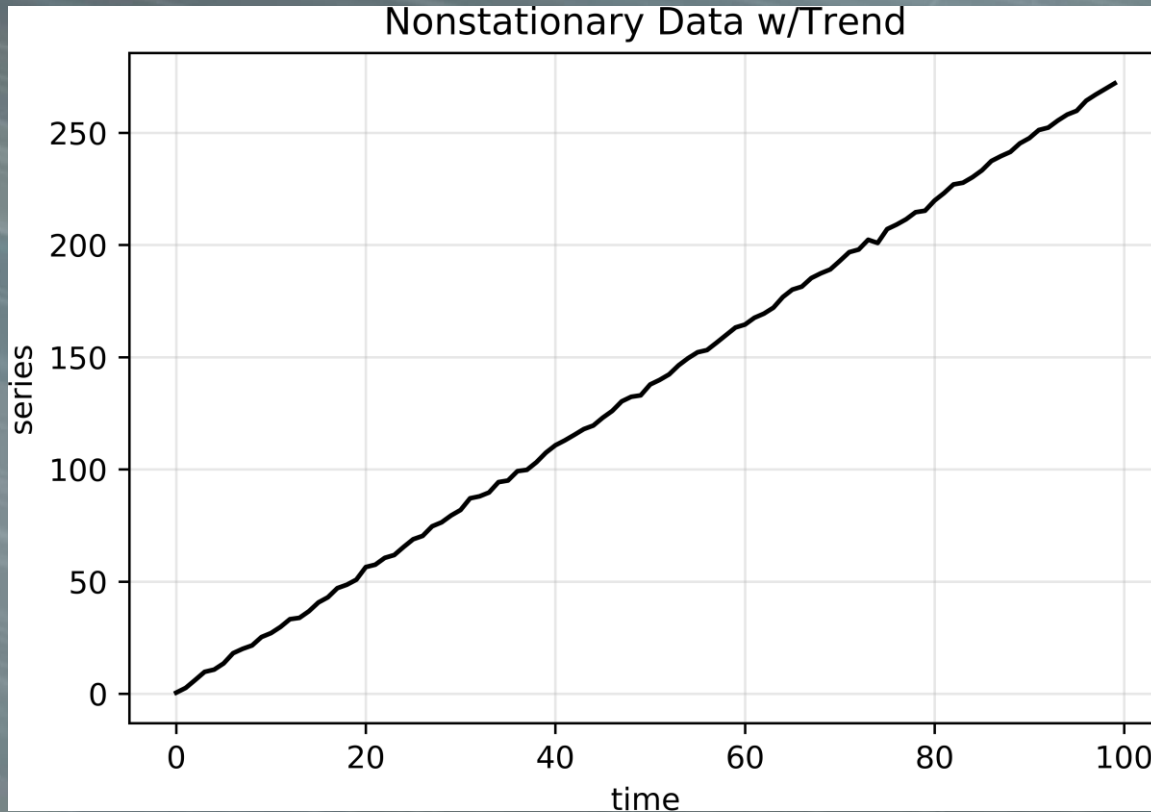
- ▶ A stationary time series has constant autocorrelation structure throughout the entire series.
- ▶ If the autocorrelation remains constant throughout the series, a simple transformation can be used to remove the autocorrelation.
- ▶ This will be useful for several future models.

Why Is Stationarity Important?

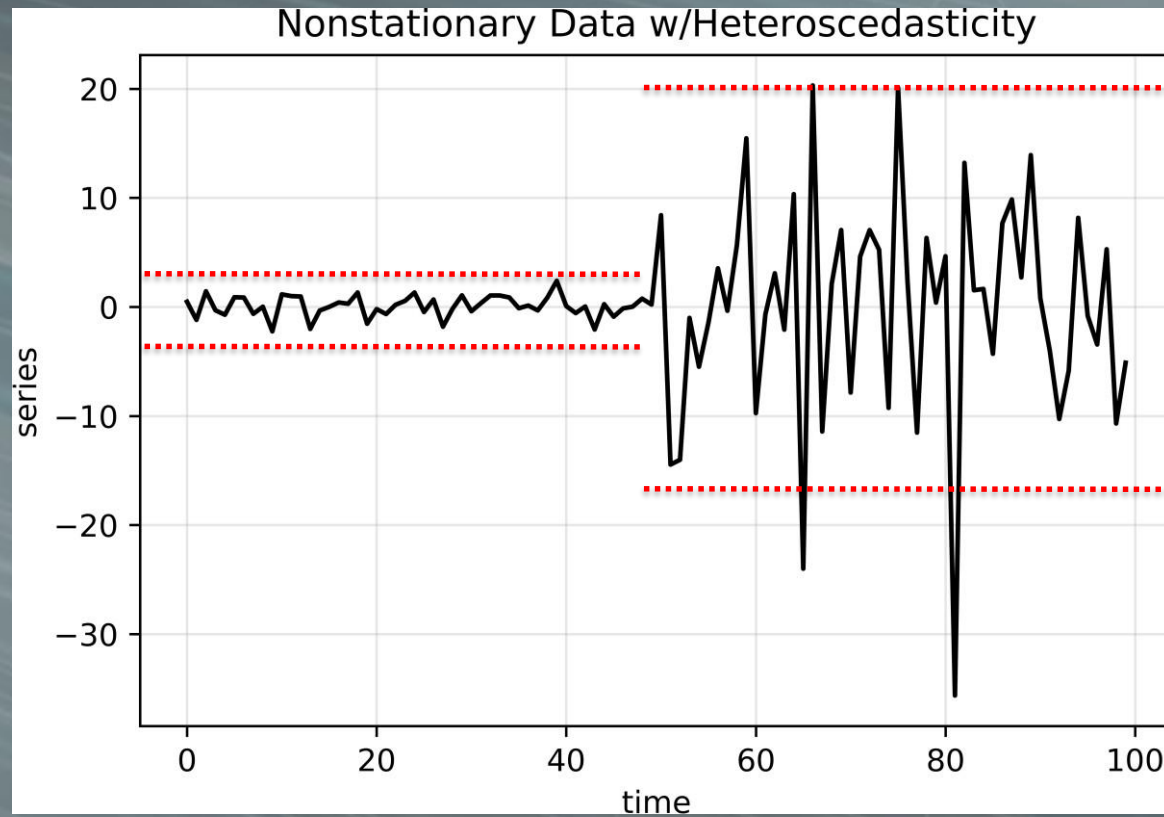
Stationarity is a fundamental assumption in many time-series forecasting models:

- ▶ Without it many basic time-series models would break down.
- ▶ Transformations can be applied to convert a nonstationary time series to a stationary one before modeling.
- ▶ While there are more advanced time-series models that can handle nonstationary data, that is beyond the scope of this lesson.

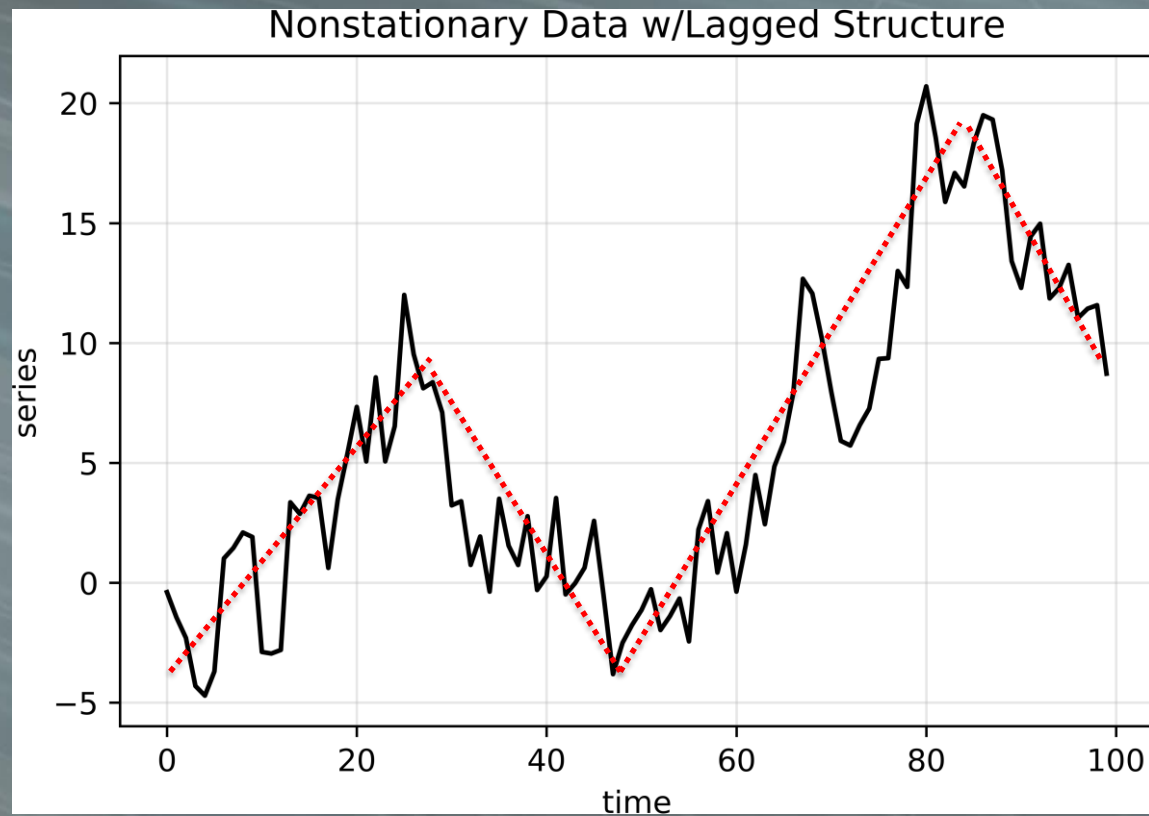
Non-stationary Example



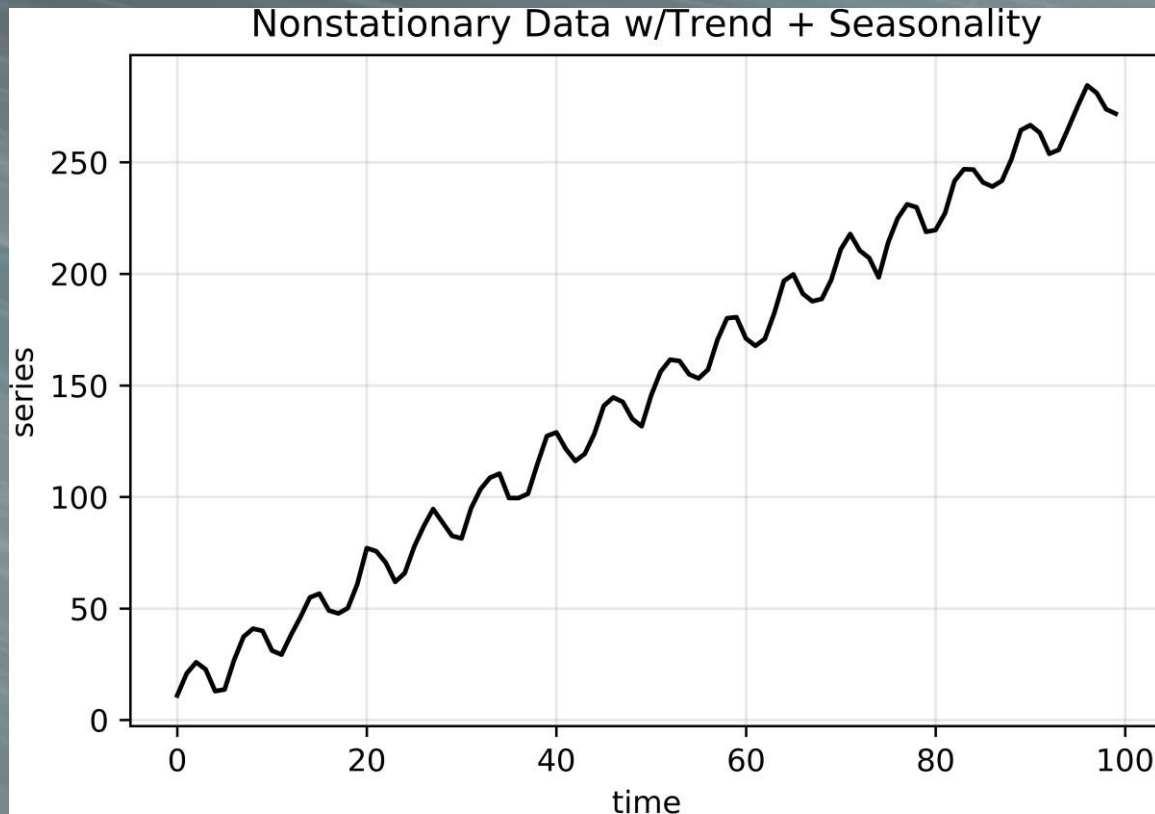
Non-stationary Example



Non-stationary Example



Non-stationary Example



Identifying Nonstationary Time-Series Data

There are several ways to identify nonstationary time-series data:

- ▶ Run-sequence plots
- ▶ Summary statistics
- ▶ Histogram plot
- ▶ Augmented Dickey-Fuller test

Run-Sequence Plot

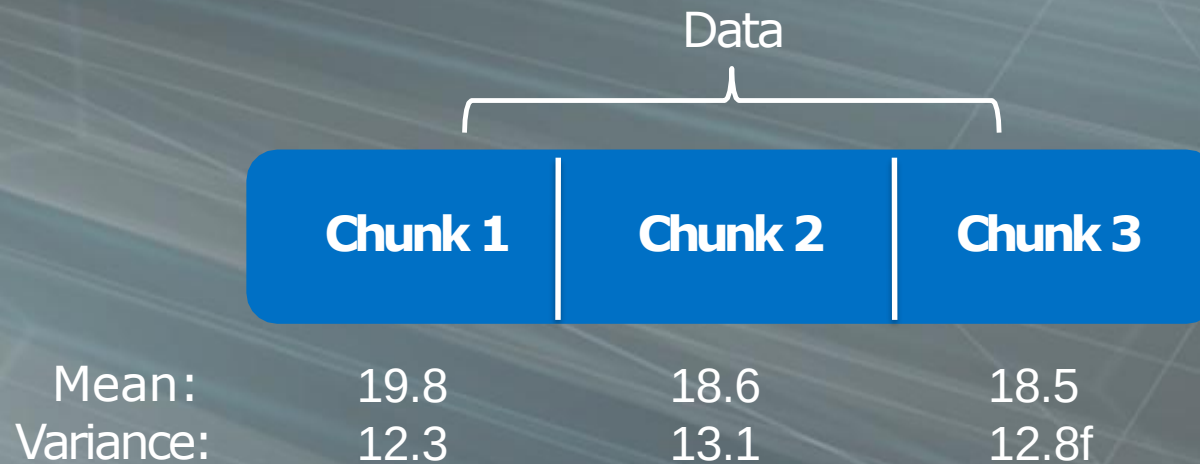
A run-sequence plot is simply a plot of your time-series data.

- ▶ This should always be your first step in time-series analysis.
- ▶ It often shows whether there is underlying structure.
- ▶ Be on the lookout for trend, seasonality, and autocorrelation.
- ▶ The previous plots are great examples.

Summary Statistics

- ▶ Calculating the mean and variance over time is a useful way to discern whether the series is stationary.
- ▶ A simple but effective way to do this is to split your data into chunks over time and compute statistics for each chunk.
- ▶ Large deviations in either the mean or the variance among chunks are problematic and mean that your data is nonstationary.

Stationary



The chunks have similar mean and variance.

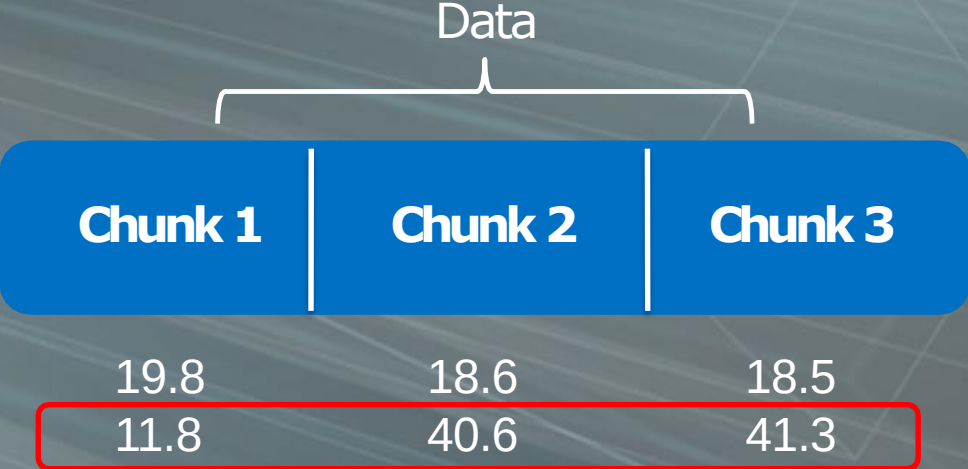
Nonstationary

The diagram illustrates a nonstationary dataset divided into three chunks. A bracket labeled 'Data' spans the three chunks. The chunks are represented by blue rounded rectangles labeled 'Chunk 1', 'Chunk 2', and 'Chunk 3'. Below these, a table shows the mean and variance for each chunk. The mean values (19.8, 30.6, 49.5) are highlighted with a red border, indicating significant variation between chunks.

	Chunk 1	Chunk 2	Chunk 3
Mean:	19.8	30.6	49.5
Variance:	12.3	13.1	12.8

There are large deviations in the mean between chunks.

Nonstationary



The diagram illustrates a nonstationary data series divided into three chunks. A bracket labeled 'Data' spans the three chunks. Below the chunks, a table lists the mean and variance for each. The variance values are highlighted with a red box to show significant differences between chunks.

	Chunk 1	Chunk 2	Chunk 3
Mean:	19.8	18.6	18.5
Variance:	11.8	40.6	41.3

There are large deviations in the variance between chunks.

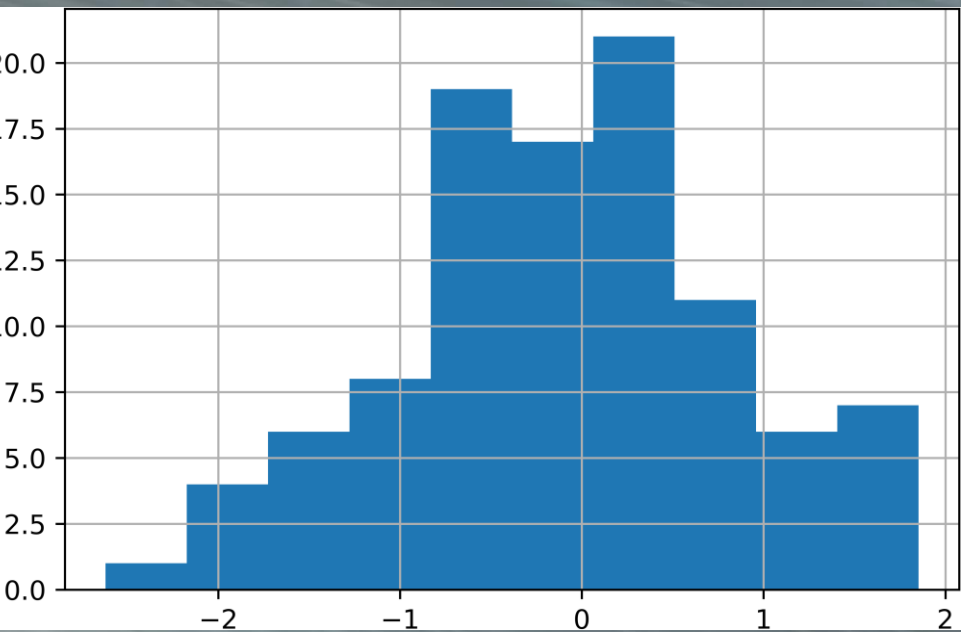
Histogram Plot

A histogram plot gives important clues into a time series' underlying structure.

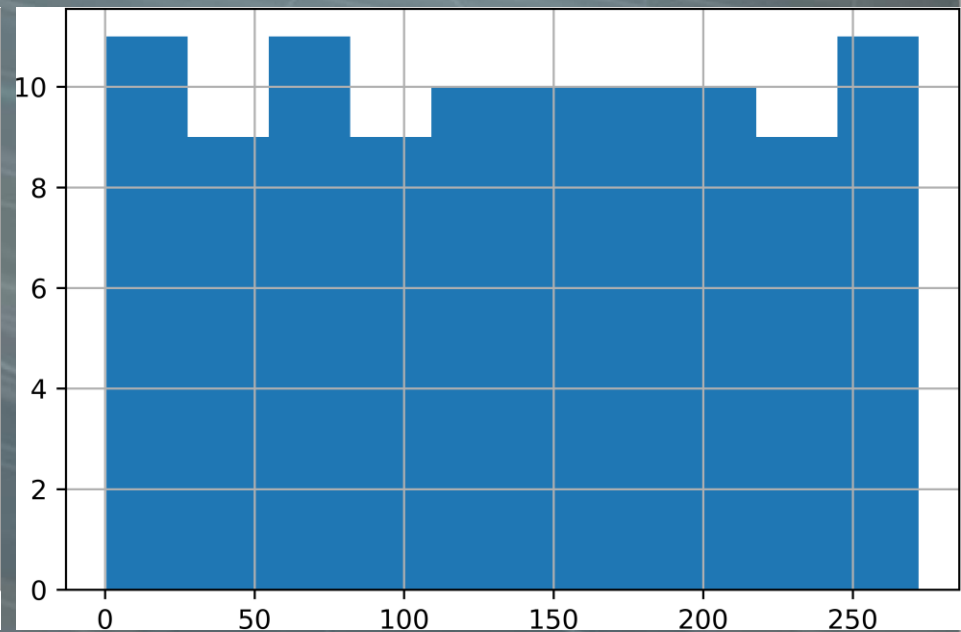
- ▶ If you see a distribution that is approximately normal, that's a good indication your time series is stationary.
- ▶ If you see a nonnormal distribution, that's a good indication your time series is nonstationary.

Histogram Plot

Stationary



Nonstationary



Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller test is a hypothesis test that tests specifically for stationarity.

- ▶ We generally say that the series is **stationary** if the p-value is less than 0.05.
- ▶ It is a less appropriate test to use with small datasets or when heteroscedasticity is present.
- ▶ It is best to pair ADF with other techniques, such as run-sequence plots, summary statistics, or histograms.

ARIMA

ARIMA Model

- ▶ Autoregressive (AR) part
- ▶ Integrated (I) part
- ▶ Moving average (MA) part

ARIMA Model Details

There are a few things you should know about ARIMA models:

- ▶ The ARIMA model is denoted **ARIMA(p, d, q)**.
 - ▶ p is the order of the AR model.
 - ▶ d is the number of times to difference the data.
 - ▶ q is the order of the MA model.
 - ▶ p, d, and q are nonnegative integers.

Correlation and Lag Factor

- ▶ Before going further, let's learn the important concepts

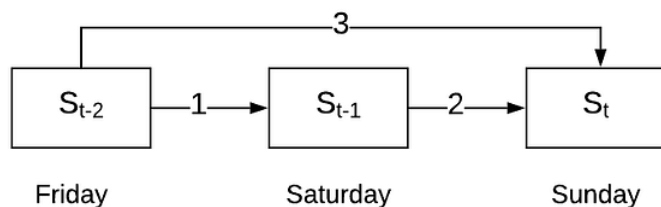
S_t = Price of Gas on Sunday

S_{t-1} = Price of Gas on Saturday

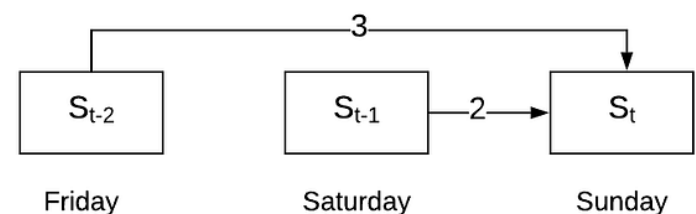
S_{t-2} = Price of Gas on Friday

Auto-correlation(AC) and Partial AC

- ▶ Auto-correlation considers both direct and indirect effects
- ▶ Partial autocorrelation consider only the direct effects



Auto-correlation



Partial Auto-correlation

Autoregressive Models (AR)

- ▶ Auto Regressive (AR) model is a specific type of regression model where, the dependent variable depends on past values of itself.

Autoregressive Models (AR)

- ▶ The notation AR(p) indicates an autoregressive model of order p

$$X_t = \sum_{i=1}^p \varphi_i X_{t-i} + \varepsilon_t$$

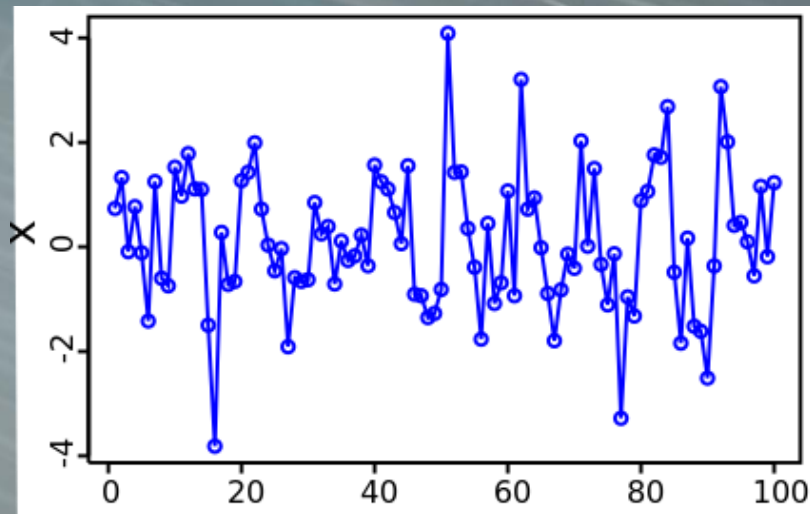
- ▶ where φ_i are the parameters of the model and ε_t is white noise (or you can think it is a constant term).

Autoregressive Models (AR)

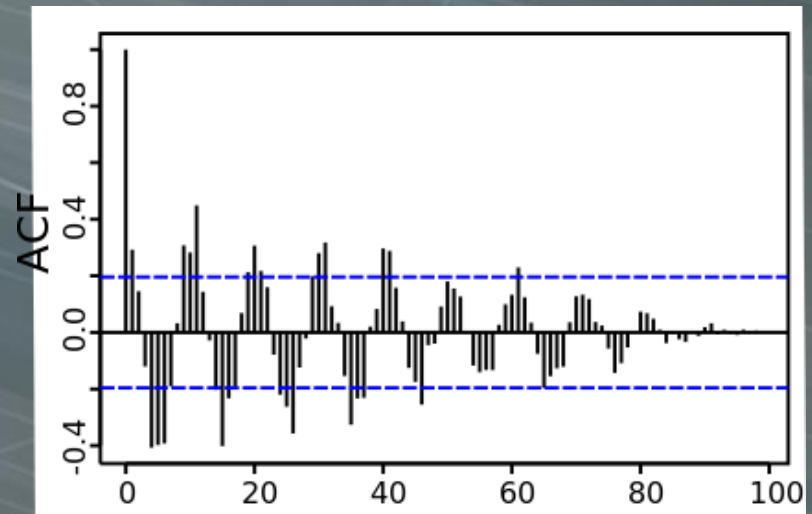
- ▶ The current observation X_t is therefore depending on previous values
- ▶ If the correlation is more, the weight corresponding to that lagged observation is high (and vice-versa).
- ▶ Type of correlation here is partial autocorrelation

ACF Plot

- ▶ We can also use Autocorrelation Function (ACF) plot to roughly check the autocorrelation relationships



Observed Data



ACF Plot

Steps for Forecasting with AR

1. Model the trend and/or seasonality of a time series and subtract it from the data ($X_t = \text{Observed}_t - \text{Trend}_t - \text{Seasonality}_t$)
2. Calculate AR on X_t to obtain X_{t+1}
3. Forecast = $X_{t+1} + \text{Trend}_t + \text{Seasonality}_t$

Autoregressive Models (AR)

- ▶ Useful for short term forecasts
- ▶ This model can forecast up to p values

Moving Average Models (MA)

- ▶ Moving Average (MA) model works by analysing how wrong you were in predicting values for the previous time-periods to make a better estimate for the current time-period.
- ▶ Moving average models are not the same as smoothing.
- ▶ Please don't confuse the two.

Moving Average Model (MA)

- ▶ The notation MA(q) indicates a moving average model of order q.

$$X_t = \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i} + \mu$$

- ▶ where θ_i are the parameters of the model, ε_t is white noise, μ is the “average” term.
- ▶ The ε terms represent the errors observed at respective lags and the weights ($\theta_1, \theta_2 \dots \theta_q$) are calculated statistically depending on the correlations.

Moving Average Models (MA)

- ▶ MA models are conceptually a linear regression of the current value of the series against the white noise of one or more of the previous values of the series.
- ▶ The noise at each point is assumed to come from a normal distribution with mean 0 and constant variance.

Intuitive Example for MA :

Daily Sales & Advertising Shocks

- ▶ A store's daily sales X_t are generally stable, but influenced by unpredicted daily shock/noise ε_t —like a surprise local news mention or a sudden rainstorm.
- ▶ Thus, the model can be :
 $(X_t = 100 + \varepsilon_t + 0.5 \varepsilon_{t-1})$
 - ▶ $\mu = 100$: The average daily sales are 100 units.
 - ▶ ε_t : Today's random event (e.g., +20 units from a sunny day).
 - ▶ $0.5 \varepsilon_{t-1}$: Yesterday's event still has a 50% "echo" effect today. If yesterday was sunny (+20), today gets a (+10) boost even if today is cloudy.

Moving Average Model (MA)

- ▶ The effects of these previous(lagged) observation errors, on the current observation, depends on the **auto-correlation** between them.
- ▶ It is possible to write any stationary AR(p) model as an MA(∞) model using repeated substitution, e.g. a AR(1) model:

$$\begin{aligned}X_t &= \phi_1 X_{t-1} + \varepsilon_t \\&= \phi_1 (\phi_1 X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \\&= \phi_1^2 X_{t-2} + \phi_1 \varepsilon_{t-1} + \varepsilon_t \\&= \phi_1^3 X_{t-3} + \phi_1^2 \varepsilon_{t-2} + \phi_1 \varepsilon_{t-1} + \varepsilon_t\end{aligned}$$

Invert the order of the terms...

.....

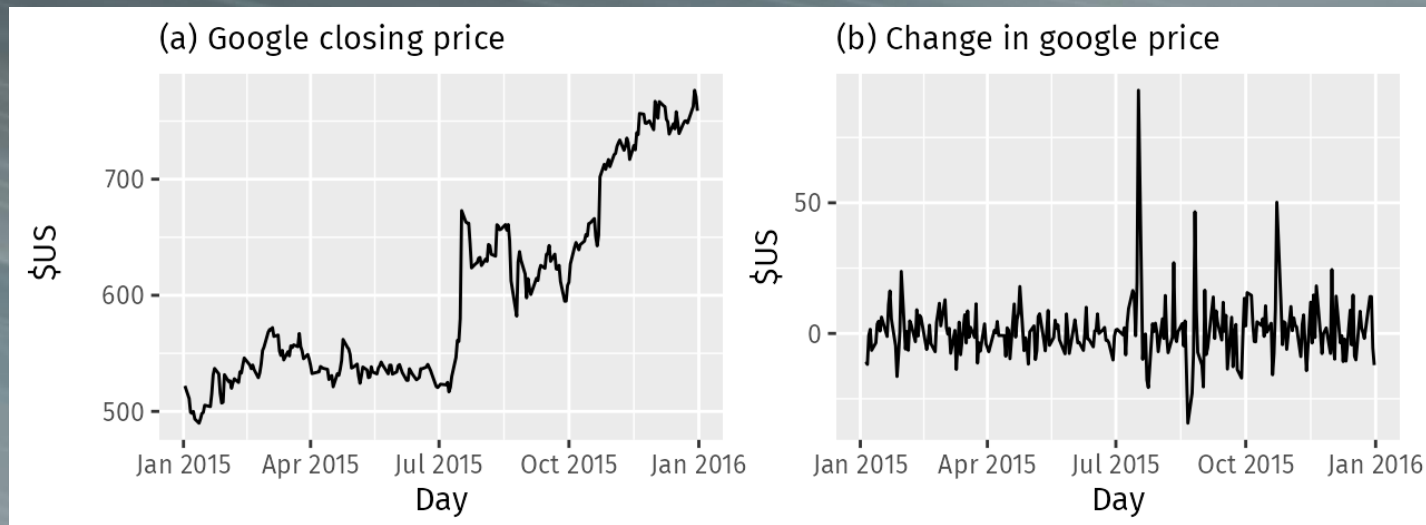
Moving Average Model (MA)

- ▶ Fitting MA estimates is more complicated than AR models because the error terms are not observable.
- ▶ As a result, iterative nonlinear fitting procedures need to be used.
- ▶ MA models are less interpretable than AR models.

Differencing

- ▶ The models of AR and MA assume that the series is stationary.
- ▶ This also means that “stationarity” is a necessary condition to take advantage of these models for any time series.
- ▶ **Differencing** can help stabilise the mean of a time series by removing changes in the level of a time series, and therefore eliminating (or reducing) trend and seasonality.

Differencing



- ▶ While ARIMA models can deal with non-stationarity up to a point, they cannot effectively account for time-varying variance.

Integrated part

- ▶ Integrated (though it has nothing to do with integration).
- ▶ means that, instead of predicting the time series itself we will predict the differences of the series from one time step to the next time step.

Order of Difference

- ▶ Apart from taking first order difference i.e. a single phase of differencing of consecutive terms. This is could done multiple times to make the series stationary.

$$Z_t = Y_{t+1} - Y_t \quad \dots d = 1$$

$$Q_t = Z_{t+1} - Z_t \quad \dots d = 2$$

...

Combined all components

- ▶ Thus, by combining AR, I and MA, the final ARIMA model will take the following form:
- ▶ The ARIMA model is denoted **ARIMA(p, d, q)**.
 - ▶ p is the order of the AR model.
 - ▶ d is the number of times to difference the data.
 - ▶ q is the order of the MA model.
 - ▶ p, d, and q are nonnegative integers.

Steps in using ARIMA model

1. Model identification
 - ▶ Check for stationarity and seasonality and perform differencing if necessary
 - ▶ Decide the order parameters
2. Model estimation
 - ▶ Finding the ARIMA terms
3. Model evaluation

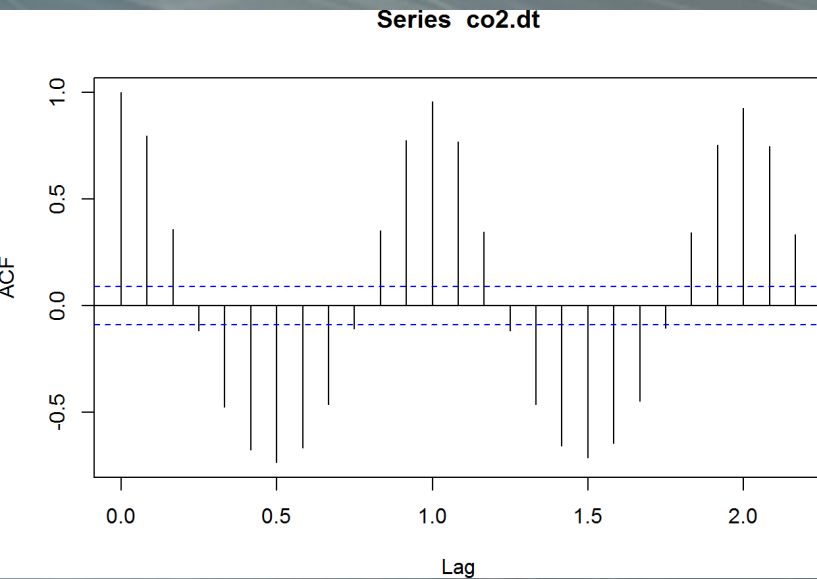
Model identification

Confirm the following:

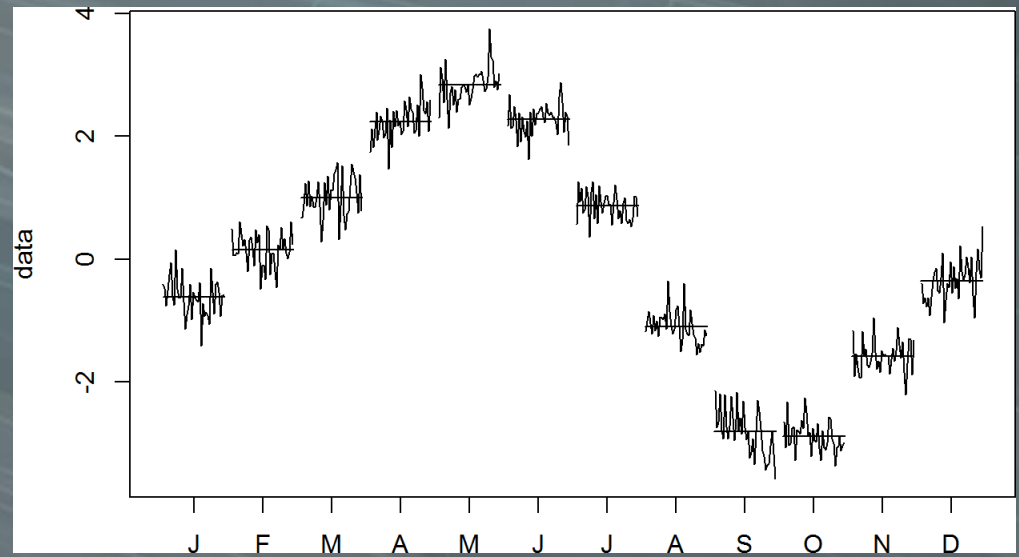
- ▶ The time series is stationary.
- ▶ Whether the time series contains a seasonal component
 - ▶ Autocorrelation plot
 - ▶ Seasonal subseries plot
 - ▶ Spectral plot

Autocorrelation plot

Autocorrelation Plot

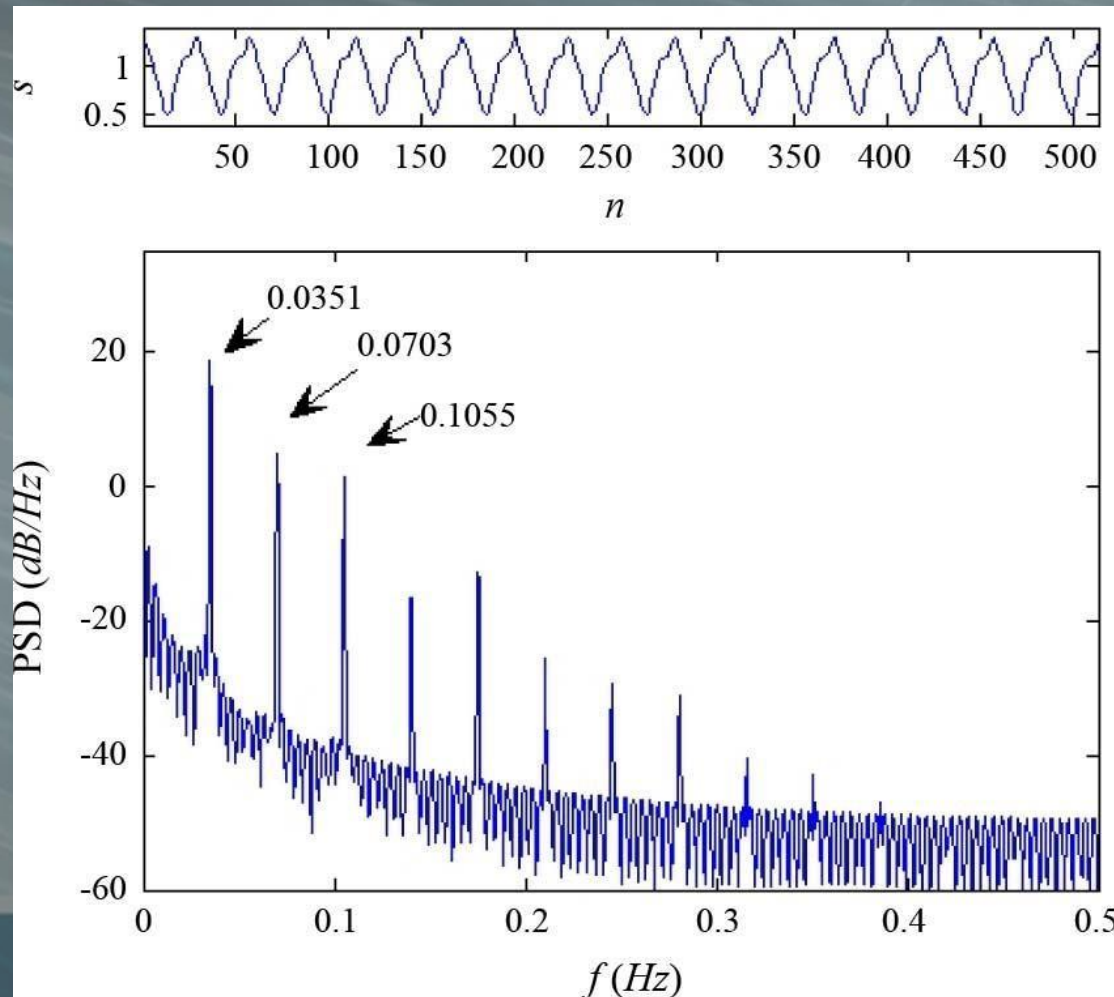


Seasonal Subseries Plot



Spectral Plot

- ▶ Make use for frequency domain transformation (e.g. Fourier Transform)



Model Identification

The first parameter: p

- ▶ the number of lagged observations.
- ▶ It determines how far back in time we go when trying to predict the current observation.
- ▶ We do this by looking at the autocorrelations of our time series, which are the correlations in our series at previous time lags.

The second parameter: d

- ▶ refers to the order of differencing. Again, differencing simply means finding the differences between consecutive timesteps.
- ▶ It is a way to make our data stationary, removing the trends and any changes in variance over time.
- ▶ d indicates differencing at which order you get a process stationary.

Model Identification

The third parameter: q

- ▶ refers to the order of the moving average (MA) part of the model.
- ▶ It represents the number of lagged forecast errors included in the model.
- ▶ MA in ARIMA captures the relationship between an observation and the residual errors in the lagged observations.

Model Evaluation

- ▶ Estimating the parameters of an ARIMA model can be a complicated, nonlinear problem.
 - ▶ Nonlinear least squares and maximum likelihood estimation (MLE) are common approaches.
 - ▶ Many modern software programs will fit the ARMA model for you

Forecasting/Error Metrics



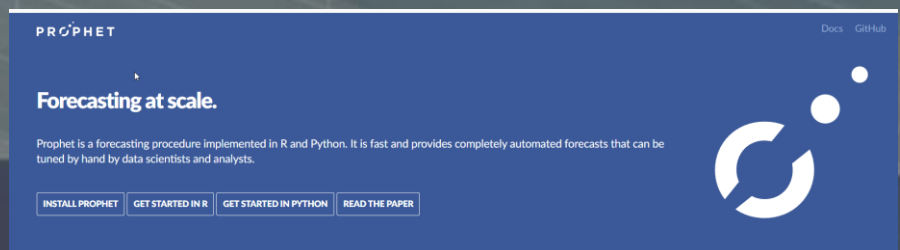
Assumptions

ARMA, ARIMA, SARIMA assumptions:

- ▶ Time-series data is stationary.
- ▶ If nonstationary, remove trend, seasonality, apply differencing, and so on.
- ▶ Remember that stationary data has no trend, seasonality, constant mean, and constant variance.
- ▶ Therefore, the past is assumed to represent what will happen in the future in a probabilistic sense. 因此，过去被视为以概率性方式预示未来可能发生的情况

Other forecasting methods/tools

- ▶ Decision tree regression
 - ▶ a predictive modeling technique that uses a tree-like model of decisions to predict the value of a target variable by learning simple decision rules inferred from the data features.
- ▶ Meta Prophet
 - ▶ based on an additive model where non-linear trends are fit with yearly, weekly, and daily seasonality, plus holiday effects. It works best with time series that have strong seasonal effects and several seasons of historical data.



Summary

- ▶ Correlation
- ▶ Regression Analysis
 - ▶ Linear Regression
 - ▶ Non-linear regression
- ▶ Time series forecasting
- ▶ Stationarity
- ▶ ARIMA